CHAPTER TWO
HTML

Web design and development

What is HTML?

- HTML is a language for describing web pages.
- HTML is the lingua franca of the web, that is it the language that makes it possible for various computers to communicate with each other.
- HTML stands for **H**yper **T**ext **M**arkup **L**anguage
- **Hypertext**: refers to the ability to create links to other pages and other web resources. (Hyperlinks: makes the web go around)
- **Markup**: means it is used for creating pages of formatted text along with images and other resources embedded in the page.

What is HTML?

- HTML is a **markup** language, not a programming language.
- A markup language is a set of markup tags.
- The tags **describe** document content
  - In other words, the markup tags provide information about the page content structure.
- HTML documents contain HTML **tags** and plain **text.**
- HTML documents are also called **web pages.**
- Originally based on SGML (Standard Generalized Markup Language).

What is HTML?...

- It is a language used to create documents for the World Wide Web.
- HTML is not **case sensitive** language.
- HTML documents can include graphics, and more importantly links to other documents – hypertext.
- An HTML document can be produced in **free-format**, i.e. it doesn't matter what the text file looks like it is the browser that does the formatting. HTML is **not** so much **concerned** about the **appearance** of the documents but about the **structure** of a document.

History of HTML

1991	HTML first published
1995	HTML 2.0
1997	HTML 3.2
1999	HTML 4.01
2000	XHTML 1.0
2002 - 2009	XHTML 2.0
2012	HTML5

After HTML 4.01 was released, focus shifted to XHTML and its stricter standards.

XHTML 2.0 had even stricter standards than 1.0, rejecting web pages that did not comply. It fell out of favor gradually and was abandoned completely in 2009.

HTML5 is much more tolerant and can handle markup from all the prior versions.

Though HTML5 was published officially in 2012, it has been in development since 2004.

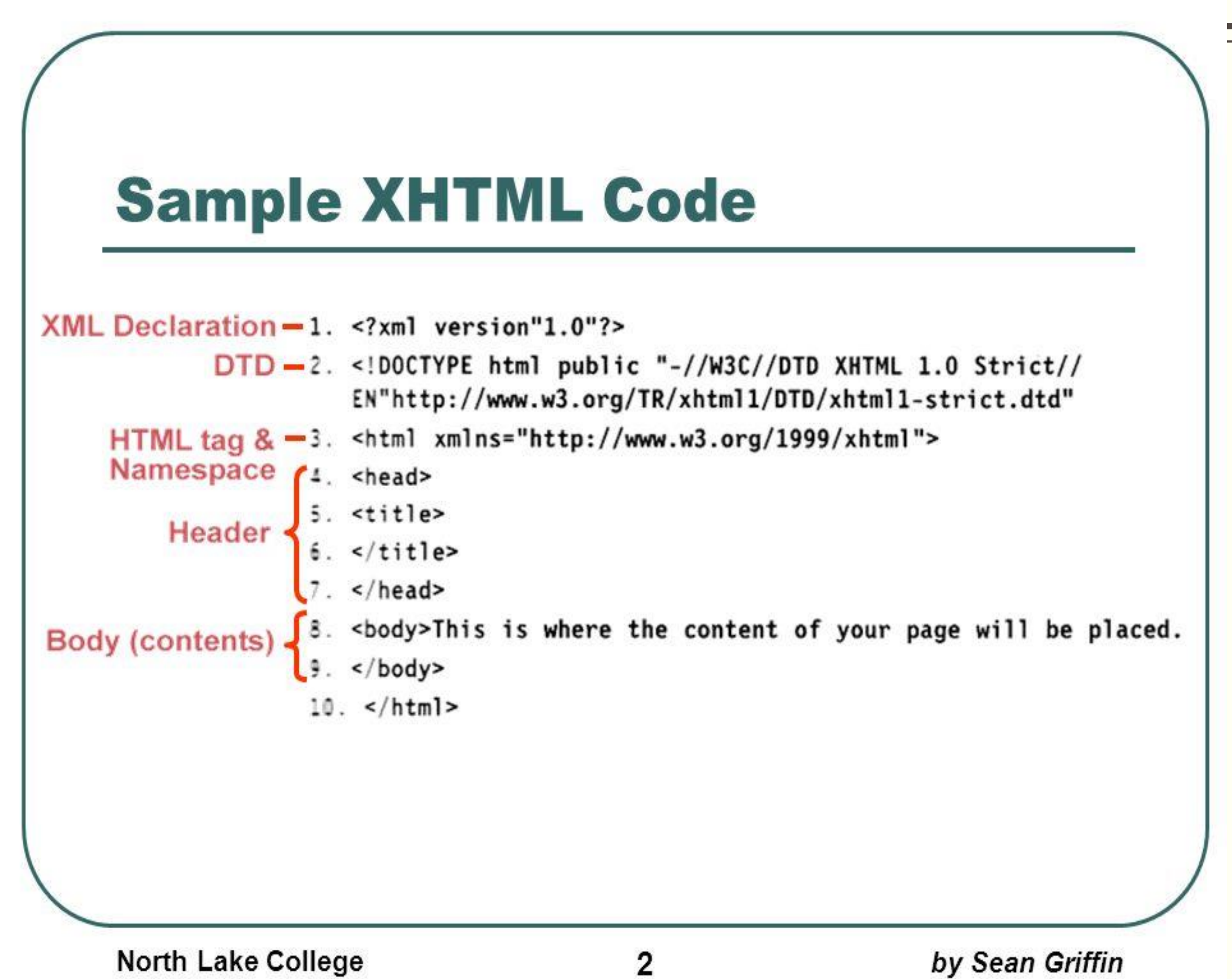
Sample xhtml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="en" lang="en">
<head>
<title> Strict DTD XHTML Example </title>
</head>
<body>
<p> Please Choose a Day: <br /><br />
<select name="day">
<option selected="selected">Monday</option> <option>Tuesday</option>
<option>Wednesday</option>
</select> </p>
</body>
</html>
```

The Birth of XHTML

- The success of XML rather left HTML in the lurch because HTML documents are not valid XML documents that the new XML processor cannot read them.
- On the other hand HTML browsers cannot read XML documents.
- An XHTML is an XML-based language that reproduces the functionality and structural rules of HTML, **but subject to the rules of XML, and uses the same mark-up elements names.** The main differences are:
  - All elements and attribute names must be in lowercase.
  - All empty elements must be written using XML's special empty-element tag
  - All end tags must be present and there are no optional end tags
  - The HTML elements must contain a single head element, followed by a single body element or single frameset element.
  - Every head must contain a single title.

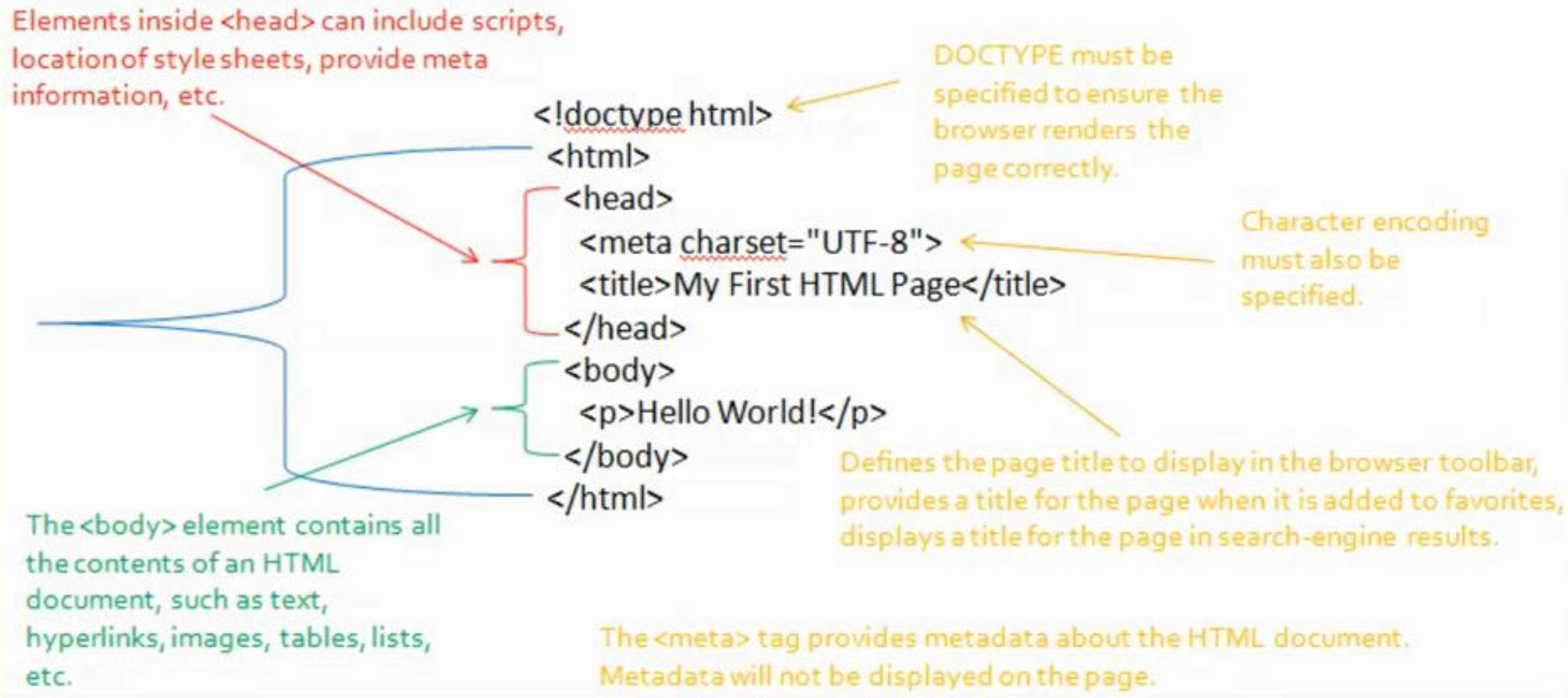
XHTML Structure



XHTML vs. HTML

- HTML is forgiving of sloppiness
- XHTML requires strict usage
- Stricter usage = better interoperability = websites work better on more devices.

HTML Page Structure



DOCTYPE Declaration

- Tell the browser what version of HTML to be expected in the document.
- **<!DOCTYPE html>**
- Should be the very first thing in your document
  - Before any comment or even a whitespace.

Structure of an HTML Document

- Every HTML document should start by declaring that it is an HTML document.
- Document Object Model tree is formed by using <html>.
- These tags are of the form:

<html>

Should appear at the beginning of your document

</html>

Should appear at the end of your document

Head of an HTML Document

- The HEAD section of an HTML document is like the front matter of the book. It where the information about the document goes. **The meta information that describes things about the document that support how it is rendered, how it is navigated, how it is assembled and otherwise how the document works.**
- The HEAD of an HTML document is where information such as the document's title, can be placed.
- The region associated with the head of a document should be declared using the following HTML tags:

  <head>

  Should appear after the <html> definition.

  </head>

  Should appear after any header text

Head of an HTML Document

- A typical entry in the HEAD of a document would be:

  <title>

  Title of the document

  </title>

Head of an HTML Document

- **Meta tag**: Used to describe various aspects of your page.
  - Search engine optimization – Now search engines are not using them
- Meta tag: data about data
  - <meta charset="UTF-8" /> - character set of the document
  - <meta http-equiv="refresh" content="3; url=http://livescore.com/" />
  - <meta http-equiv="content-type" content="text/html; charset=UTF-8" />
  - <meta name="keywords" content="Amazing,New,Bill,Page,blah blah" />

Quiz 1 ..Using Meta tag

Create HTML 5 document by the name Quiz.html with the following meta tags?

- Description
- Keywords
- Robots
- Redirect meta tag which redirects to another html document after 5 second.
- Add a base name to your document
- And also create an icon

Hint

- **<meta name=“” content=“” />**
- **<base href=“” />**
- **<link rel=“icon” href=“” />**

Body of an HTML Document

- The BODY of an HTML document is where all the information you wish to view must appear.
- The region associated with the BODY of a document should be declared using the following HTML tags:

  <body>

  Should appear after the </head> definition

  </body>

  Should appear after the document's text but before the </html> tag
- An HTML document can be freely formatted as you wish.

Web Browsers

- The purpose of a web browser (such as Google Chrome, Internet Explorer, Firefox, Safari) is to read HTML documents and display them as web pages.
- The browser does not display the HTML tags, but uses the tags to determine how the content of the HTML page is to be presented / displayed to the user.
- In other words, the **purpose** of a web browser (like Internet Explorer or Firefox) is to read **html documents** and **display** them as **web pages**.

Creating HTML Pages

- An HTML file must have an .htm or .html file extension.
- HTML files can be created with text editors:
  - NotePad, Note Pad ++, Visual Studio Code
- Or HTML editors
  - Microsoft FrontPage
  - Macromedia Dreamweaver
  - Netscape Composer
  - Expression Web
  - Visual Studio

HTML Tags

- HTML markup tags are usually called HTML tags.
- HTML tags are keywords (tag names) surrounded by **angle brackets** like <html>
- HTML tags normally **come in pairs** like and
- The first tag in a pair is the **start tag**, the second tag is the **end tag**
- The end tag is written like the start tag, with a **forward slash** before the tag name
- Start and end tags are also called **opening tags** and **closing tags**
- <tagname>content</tagname>
- Follows the <tag_on> </tag_off> format

HTML Tags…

- HTML tags should always be nested and never over-lapping
- Most HTML tags are like brackets – they form pairs; and various pairs must always match.
- For example, the brackets:

    [(< and >)] match;

    whereas the grouping:

    [(< and)]> form a mis-match!

HTML Elements

- “HTML tags” and “HTML elements” are often used to describe the same thing.
- But strictly speaking, an HTML element is everything between the start tag and the end tag, including the tags:
- HTML Element:

<p>This is a paragraph.</p>

Types of HTML Elements

- There are two different types of tags:
  - **Container Element:** Container Tags contains start tag & end tag
    i.e. <html>…</html>
  - **Empty Element:** Empty Tags contains start tag
    i.e.
    Better written with / as

HTML Attributes

- HTML elements can have attributes
- Attributes provide **additional information** about an element
- Attributes are always specified in **the start tag**
- Attributes come in name/value pairs like: **name="value"**
- The quotes for value in attributes are optional, in XHTML it is required.
- Can use either a single quote or a double quote.

Content Models in HTML5

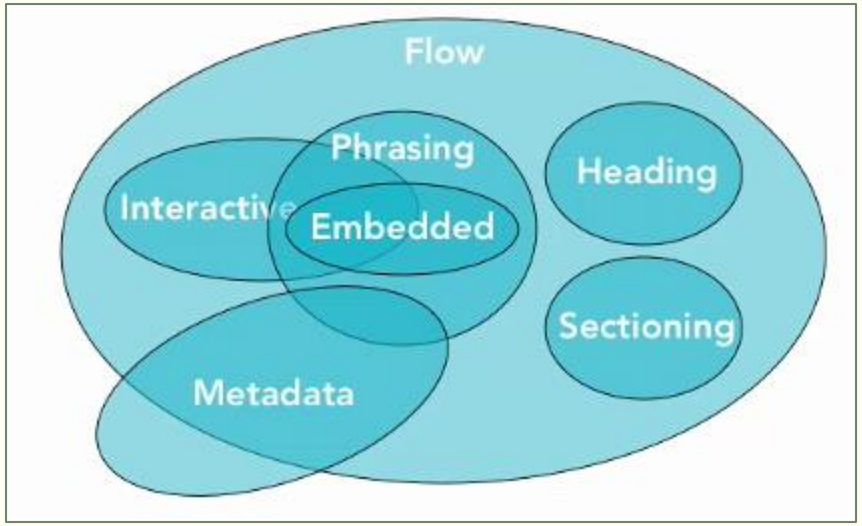
A description of the element's expected contents

- Previous version of HTML
  - Block and Inline
- A content model defines **what types of content are excepted in certain context**. A given element may only have content that matches the requirements of its content model.
- **A description of the element's expected contents.**
- An element’s content model may include zero or more categories of content and an element may also belong to zero or more categories.

Content Models in HTML5

A description of the element's expected contents

- In HTML5 there are seven content models. These content models overlap each other that is some elements may belong to more than one content model.



Content Models in HTML5

A description of the element's expected contents

- **Flow:** Most elements that are used in a body of an HTML document are flow content. These includes text, paragraphs, headings, lists, hyperlinks images, embedded media… just about everything.
- **Phrasing:** includes the text of the document as well as elements that markup the text. These includes images, hyperlinks, form elements, text markup like <b> and <i> elements and pretty much anything else you can put in a paragraph.
- **Heading:** includes <h1> through <h6> heading elements and the new <hgroup> element.
- **Sectioning:** includes the new sectioning elements <article> <aside> <nav> <section>

Content Models in HTML5

A description of the element's expected contents

- **Embedded:** is a content that imports another resources into the document or content that is otherwise inserted into the document. Includes video, audio, flash other plugins as well as images, canvas and even math object.
- **Interactive:** is a content that is specifically intended for user interaction. Like hyperlinks, form elements, menus and even media if it has the controls attribute.
- **Metadata:** is a content that sets up the presentation or behavior of the rest of the content or that sets up the relationship of the document with other documents or conveys other out of bound information.

- Read More*:- https://www.w3.org/TR/2011/WD-html5-20110525/content-models.html*

Quiz 2

- Classify the following elements based on the 7 content models ?

1. <a>
2. <p>
3. <audio>
4. <b><i>
5. <article>

Whitespace in HTML

```html
<p>
  This is a
              line     of
      text. </p>



<p> This is another line of text. </p>
```

This is a line of text.

This is another line of text.

- By using the special element pre, we can display whitespaces.

```html
<pre>
  This is a
              line     of
      text. </pre>



<p> This is another line of text. </p>
```

```
  This is a
              line     of
      text.
```

This is another line of text.

Comments in HTML

- Comments can exists anywhere between the

  <html></html> tags

- Comments start with <!-– and end with -->

```
<!– Company Logo (a JPG file) -->
<img src="logo.jpg" alt="Company Logo">
<!–- Hyperlink to the web site -->
<a href="http://Microsoft.com/">Microsoft</a>
<!–- Show the news table -->
<table class="newstable">
…
```

Display Modes
> Block and Inline Elements (tags)

- **Block elements:**
  - Elements that contain other elements.
  - Add a line break before and after them, and expand to 100% width.
  - **Eg. <div>, <p>, <h1>, <ul> are block elements**
  - <p> tag is the only block level element that cannot contain other block level elements. Can only contain text and inline level elements.
- **Inline elements:**
  - Typically represent the current inline of text.
  - Don’t break the text before and after them, they flow with it.
  - **Eg. <span>, <a>, <em>, <img> are inline elements**

Block and Inline Elements

- Can change inline elements to block using stylesheet.
  - style="display: block"
- Block elements can also be changed to inline elements.
  - style="display: inline"

HTML Paragraphs

- HTML will treat all text as one paragraph unless paragraph markers are added around each physical paragraph.
- HTML documents are divided into paragraphs.
- HTML paragraphs are defined with the <p> tag.
- One can also write texts outside of a paragraph but it is always recommended to write a text inside a paragraph or another block element.

```
<p>This is a paragraph</p>

<p>This is another paragraph</p>
```

This is a paragraph

This is another paragraph

HTML Paragraphs… > Alignment

- Paragraphs by default are aligned to the left.

A paragraph is typically formatted as a sequence
words, wrapped at the margins, and with a blank
separating the paragraph from other paragraphs o
the page.

```
<p>
    A paragraph is typically formatted as a sequence of words,
    wrapped at the margins, and with a blank line separating the
    paragraph from other paragraphs on the page.
</p>
```

- You may use the align attribute to align paragraphs but it is not recommended and it is actually obsolete on the latest version of HTML.

HTML Paragraphs

- A better way to do this is using CSS.

```
<p style="text-align: right">
    A paragraph is typically formatted as a sequence of words,
    wrapped at the margins, and with a blank line separating the
    paragraph from other paragraphs on the page.
</p>
```

A paragraph is typically formatted as a sequence of
words, wrapped at the margins, and with a blank line
separating the paragraph from other paragraphs on
the page.

Character References

Character	Named	Decimal	Hexadecimal
<	<	<	<
>	>	>	>
&	&	&	&
“	"	"	"
‘	'	'	'

- http://developers.whatwg.org/named-character-references.html

Line Break Tag

- If you want a line break or a new line without starting a new paragraph, use the
tag.

```
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. <br />Duis quis
    porta nunc. Aliquam quis nulla turpis. Nam tincidunt nunc hendrerit
    ligula aliquet vel feugiat diam pharetra.
</p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit.
Duis quis porta nunc. Aliquam quis nulla turpis. Nam tincidunt
nunc hendrerit ligula aliquet vel feugiat diam pharetra.

Line Break and Spaces
> Prevent default line break / wrap

- The browser also controls the spacing between words.   (special character) – non-braking space

```
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit.
        Duis quis
    porta nunc. Aliquam quis nulla turpis. Nam tincidunt nunc hendrerit
    ligula aliquet vel feugiat diam pharetra.
</p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit.   Duis
quis porta nunc. Aliquam quis nulla turpis. Nam tincidunt nunc
hendrerit ligula aliquet vel feugiat diam pharetra.

Phrase markup elements

- They are designed to markup parts of text according to their usage.
- Let non visual browsers to know what to do with your text.
- All phrase markup elements are inline markup elements.
- <em>: Defines emphasized text
- <strong>: Defines important text
- <q>: Defines a quoted text (is not supported on all browsers)
- <code>: Defines monospace text
- <abbr>: Defines abbreviation
- <acronym>: Defines acronym (meaning)
- <dfn>: Defines instance (define a word when you use it for the first time)

Phrase markup elements

- <kdb>: Defines a keyboard element
- <sample>: Defines a sample text
- <var>: Defines variable
- <sub>: Defines a subscript
- <sup>: Defines a superscript

Phrase markup Elements
> Example

```html
<p> <em> Emphasize </em> </p>
<p> <strong> Strong </strong> </p>
<p> <dfn> Define a term </dfn> </p>
<p> <code> Code listings </code> </p>
<p> <samp> Sample </samp> </p>
<p> <kbd> Keyboard </kbd> </p>
<p> <var> Variables </var> </p>
```

Emphasize

Strong

Define a term

`Code listings`

`Sample`

`Keyboard`

Variables

Font markup elements

- Different from phrase markup elements.
- Defines the visual style that the elements are going to be rendered on a visual browser on the screen.
- The phrase markup elements are for giving meaning to the text.
- In reality, if you care about how your text is going to be rendered on the screen you want to use the font level markups not the phrase level markups.
- Phrase level markups don’t guarantee how the phrase is going to be rendered on the screen.

Font markup elements

- Inline markup elements and they are container elements.
- <b>: Defines bold text
- <i>: Defines a part of text in an alternate voice or mood, italic
- <big>: Defines big text
- <s> or <strike>: Defines strike text (Strikethrough)
- <small>: Defines smaller text
- <tt>: Defines teletype text (renders it in monospace font)
- <u>: Defines underline text

Phrase & Font markup elements
> Example

```html
<p> <em> Emphasize </em> </p>
<p> <strong> Strong </strong> </p>
<p> <dfn> Define a term </dfn> </p>
<p> <code> Code listings </code> </p>
<p> <samp> Sample </samp> </p>
<p> <kbd> Keyboard </kbd> </p>
<p> <var> Variables </var> </p>
```

```html
<b>This text is Bold</b>
<br><em>This text is Emphasized</em>
<br><i>This text is Italic</i>
<br><small>This text is Small</small>
<br>This is<sub> Subscript</sub> and <sup>Superscript</sup>
<br><strong>This text is Strong</strong>
<br><big>This text is Big</big>
<br><u>This text is Underline</u>
<br><strike>This text is Strike</strike>
<br><tt>This text is Teletype</tt>
```

This text is Bold
This text is Emphasized
This text is Italic
This text is Small
This is $_{\text{Subscript}}$ and $^{\text{Superscript}}$
This text is Strong
This text is Big
This text is Underline
~~This text is Strike~~
`This text is Teletype`

Mark Tag

- HTML provides a new inline element called mark for highlighting text. (<mark>....</mark>)
- Can be styled with stylesheet.

```
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis
    porta nunc. <mark>Aliquam quis nulla turpis. </mark>Nam tincidunt nunc hendrerit
    ligula aliquet vel feugiat diam pharetra.
</p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis porta nunc. Aliquam quis nulla turpis. Nam tincidunt nunc hendrerit ligula aliquet vel feugiat diam pharetra.

HTML Headings

- Headings in HTML add structure to your document.
- **Valuable for search engines and other automated tools that might want to categorize your documents** and to understand the structure of them a little bit so that they can give you intelligent results.
- A heading level is declared by using the following tags:
  - <hnumber>
    - Should appear at the beginning of your heading.
  - </hnumber>
    - Should appear at the end of your heading.

Accessibility Tip 💡

HEADING BEST PRACTICES

- Keep only one <h1> element on a web page.
- Always write <h1> - <h6> elements in hierarchical order.
- Do not skip heading levels. That means, do not write <h6> directly after <h2>.

CODE WITH SHRIPAL

Why?

It helps screen reader users in:

- Understanding the content structure of the web page.
- Navigating to a particular section of the web page quickly.

HTML Headings

- HTML headings are defined by the <h1> to <h6> tags. (Six heading elements)
- All the six heading elements are container tag and requires a closing tag.
- <h1> will print the largest heading.
- <h6> will print the smallest heading.
- It is a block level element and other block elements can be nested within a heading tag.
- Example:

```
<h3> Heading 3
        <p> paragraph in heading </p>
</h3>
```

HTML Headings…

- Example

  <h1>This is a heading</h1>

  <h2>This is a heading</h2>

  <h3>This is a heading</h3>

- It is very important that you balance your text to the various heading styles. In most cases heading level 1 should be used for top level or main headings and heading level 2 and 3 for “side” headings.

- Note: **Heading level 4 is the same size as the default font and higher levels of heading begin to get smaller in comparison to the body text.**

HTML Headings...

```html
<h1> Heading 1 </h1>
<h2> Heading 2 </h2>
<h3> Heading 3 </h3>
<h4> Heading 4 </h4>
<h5> Heading 5 </h5>
<h6> Heading 6 </h6>
```

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

Quotes

- Quotation marks can of course be added using keyboards. But smarter quotes can be added using the <q> element.
- <q>: might not be supported by all browsers but it adds a double quote around some text.
- It also uses the smart rule if it is nested.
- Different for different browsers unless a stylesheet is written.

```html
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis
    porta nunc. <q>Aliquam quis nulla turpis.</q> Nam tincidunt nunc hendrerit
    ligula aliquet vel feugiat diam pharetra.
</p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis porta nunc. “Aliquam quis nulla turpis.” Nam tincidunt nunc hendrerit ligula aliquet vel feugiat diam pharetra.

Quotes

- Block Quote: It is a block level quote.
- <blockquote>: doesn’t put a quote around a text, but adds some indenting.

```
<blockquote cite="http://www.worldwildlife.org/who/index.html">
For 50 years, WWF has been protecting the future of nature. The world's leading
conservation organization, WWF works in 100 countries and is supported by 1.2 million
members in the United States and close to 5 million globally.
</blockquote>
```

```
<blockquote>
  This is a blockquote. It can be used to quote a passage from another source.
</blockquote>
```

Preformatted Text

- Sometime you might want to format your text the way you see it on your text editor.
- The <pre> tag allows you to achieve that.
- It is a block level tag.
- Text is mono spaced.
- Inline formatting is supported inside it.
- Used to display a code (programming code) or some poetry on a webpage.

```
<pre>
          This text is
        preformatted
     and should  be
   displayed in a
     particular way
       and shape without any
         reformatting
       by
     the
   browser.

     The
       format
           is the
                poetry.
</pre>
```

```
          This text is
        preformatted
     and should  be
   displayed in a
     particular way
       and shape without any
         reformatting
       by
     the
   browser.

     The
       format
           is the
                poetry.
```

Preformatted Text

```
<pre>
int main( int argc, char ** argv ) {
    printf(&quot;Hello, World!\n&quot;);
    return 0;
}
</pre>

<pre>
&lt;html&gt;
&lt;head&gt;
    &lt;meta charset=&quot;UTF-8&quot; /&gt;
    &lt;title&gt;
        HTML Preformatted Text
    &lt;/title&gt;
&lt;/head&gt;
&lt;body&gt;
&lt;h1&gt;
    HTML Preformatted Text
&lt;/h1&gt;
&lt;p&gt;
    Hello, World!
&lt;/p&gt;
&lt;/body&gt;
&lt;/html&gt;
</pre>
```

```
int main( int argc, char ** argv ) {
    printf("Hello, World!\n");
    return 0;
}

<html>
<head>
    <meta charset="UTF-8" />
    <title>
        HTML Preformatted Text
    </title>
</head>
<body>
<h1>
    HTML Preformatted Text
</h1>
<p>
    Hello, World!
</p>
</body>
</html>
```

Quiz 3 ..Create the following page **quiz3.html**

Use pre and code

```
let mySample=()=>{

  console.log('this is a sample function');
}
```

Use q and blockquote

"this is whhat abebe is saying"

> Each element defined in this specification has a content model: a description of the element's expected content
> from https://www.w3.org/content-models.html

HTML List

- Lists provide methods to show item or element sequences in document content. There are three main types of list:
  - Unordered Lists: unordered list are bulleted
  - Order lists: Ordered lists are numbered
  - Definition lists: used to create a definition list

Unordered List

- An unordered list is a list of items. The list items are marked with bullets (typically small black circles).
- It is a block level element
- An unordered list starts with the <ul> tag. Each list item starts with the <li> tag.
- Different kinds of bullets:
  - disc (default)
  - circle
  - square

    <ul Type="disc">… (obsolete)
- Can nest one unordered list inside another unordered list

Unordered List :

- ThinkPositve
- Never Depressed
- Keep Smiling

Unordered List…
> Example

```
<h4>Disc bullets list:</h4>
<ul style="list-style-type: disc">
    <li>Jones</li>
    <li>Smith</li>
    <li>Hayes</li>
    <li>Jackson</li>
</ul>
<h4>Circle bullets list:</h4>
<ul style="list-style-type: circle">
    <li>Jones</li>
    <li>Simth</li>
    <li>Hayes</li>
    <li>Jackson</li>
</ul>
<h4>Square bullets list:</h4>
<ul style="list-style-type: circle">
    <li>Jones</li>
    <li>Smith</li>
    <li>Hayes</li>
    <li>Jackson</li>
</ul>
```

Disc bullets list:

- Jones
- Smith
- Hayes
- Jackson

Circle bullets list:

- Jones
- Simth
- Hayes
- Jackson

Square bullets list:

- Jones
- Smith
- Hayes
- Jackson

Unordered List…
> Example

```html
<ul>
    <li>One</li>
    <li>Two
        <ul>
            <li>this item
                <ul>
                    <li>this item</li>
                    <li>that item</li>
                    <li>the other item</li>
                </ul>
            </li>
            <li>that item</li>
            <li>the other item</li>
        </ul>
    </li>
    <li>Three</li>
</ul>
```

- One
- Two
    - this item
        - this item
        - that item
        - the other item
    - that item
    - the other item
- Three

Ordered List

- An ordered list is also a list of items. The list items are marked with numbers.
- It is a block level tag.
- An ordered list starts with the <ol> tag. Each list item starts with the <li> tag.
- The TYPE attribute has the following possible values, but not limited to:
  - type = "1" (Arabic numbers) or type="number"
  - type = "a" (Lowercase alphanumeric) or type="lower-alpha"
  - type = "A" (Uppercase alphanumeric) or type="upper-alpha"
  - type = "i" (Lowercase Roman numbers) or type="lower-roman"
  - type = "I" (Uppercase Roman numbers) – or type="upper-roman"
- By default Arabic numbers are used.
- You can nest an ordered list inside another ordered list.

Ordered List…

- The start attribute for ordered list allows you in specifying from what number your list should start.

Ordered List > Example

```html
<h4>Numbered list:</h4>
<ol>
    <li>Jones</li>
    <li>Smith</li>
    <li>Hayes</li>
    <li>Jackson</li>
</ol>
<h4>Letters list:</h4>
<ol type="A">
    <li>Jones</li>
    <li>Smith</li>
    <li>Hayes</li>
    <li>Jackson</li>
</ol>
<h4>Roman numbers list:</h4>
<ol type="I">
    <li>Jones</li>
    <li>Smith</li>
    <li>Hayes</li>
    <li>Jackson</li>
</ol>
```

Numbered list:

1. Jones
2. Smith
3. Hayes
4. Jackson

Letters list:

A. Jones
B. Smith
C. Hayes
D. Jackson

Roman numbers list:

I. Jones
II. Smith
III. Hayes
IV. Jackson

Ordered List…
> Example

```html
<ol type="number">
    <li>One</li>
    <li>
        Two
        <ol type="I">
            <li>aye</li>
            <li>bee</li>
            <li>see</li>
        </ol>
    </li>
    <li>Three</li>
    <li>Four</li>
</ol>
```

1. One
2. Two
    I. aye
    II. bee
    III. see
3. Three
4. Four

Definition List / Description List

- A definition list is not a list of single items. It is a list of items (terms), with a description of each item (term).
- A definition list starts with a <dl> tag (definition list).
- Each term starts with a <dt> tag (definition term).
- Each description starts with a <dd> tag (definition description).
- Definition is indented.
- Renders without bullets.

Definition List :

Success
    Fail First,
Happy
    Smile Always

Definition List
> Example

```
<dl>
    <dt>One</dt>
    <dd>The first non-zero number.</dd>

    <dt>Two</dt>
    <dd>The number after one.</dd>

    <dt>Three</dt>
    <dd>To get ready.</dd>

    <dt>Four</dt>
    <dd>Now Go, Cat, Go!</dd>

</dl>
```

One

    The first non-zero number.

Two

    The number after one.

Three

    To get ready.

Four

    Now Go, Cat, Go!

Suggesting word-break opportunities

- HTML provides two techniques for breaking words.
  - The soft hyphen ()
  - The <wbr> tag

```
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis
    porta nunc. Aliquamquisnullaturpis nam tincidunt nunc hendrerit
    ligula aliquet vel feugiat diam pharetra.
</p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis porta nunc. Aliquamquisnullaturpis
nam tincidunt nunc hendrerit ligula aliquet vel feugiat diam pharetra.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis porta nunc.
Aliquamquisnullaturpis nam tincidunt nunc hendrerit ligula aliquet vel feugiat diam pharetra.

Suggesting word-break opportunities

- Soft hyphen (named entity:)

```
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis
    porta nunc. Ali&shy;qua&shy;mqui&shy;snull&shy;atur&shy;pis nam tincidunt nunc hendrerit
    ligula aliquet vel feugiat diam pharetra.
</p>
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis quis porta nunc. Aliquamqui-
snullaturpis nam tincidunt nunc hendrerit ligula aliquet vel feugiat diam pharetra.

- Alternatively one can use <wbr/> inplace of the soft hyphen ().
  - <wbr /> does the same wrapping, but it does it without a hyphen.
- Either of these tags, disrupt text to speech applications.

Quiz5.html

morning drinks

- tea
- coffee
    1. hot
    2. cold
- milk

using defination list

---

html5
    html5 is a markup language
CSS
    Cascading style sheets
js
    java script

HTML Images

- Images are a major part of most website design.
- While the web is not exclusively a visual medium it is primarily so and images are a valuable parts of that equation.
- HTML Image is an **inline tag**.
- HTML images are defined with the <img> tag.
- To display an image on a page, you need to use the **src** attribute.
- Src stands for “source”. The value of the src attribute is the URL of the image you want to display on your page.

HTML Images…

- Common image formats:
  - JPEG which is used for photographs
  - GIF which is used for solid color images such as icons and logos
  - PNG which provides versatility but may not be supported by older browsers.
- It is an empty tag.

- Example:

HTML Images…
> Image Attributes

- **src** Defines an image on a page, src stands for source (can be relative or absolute)
- **alt** Define “alternate text” for an image

  Required in the specification

  Provides text that will be displayed on the event the browser can’t display the image, or if the browser is able to load the image but it takes a while to get around to it.

  For non visual browsers.
- **title** Defines the text displayed when you hover the mouse over the image, a tooltip
- **width** Defines the width of the image (not required but important)
- **height** Defines the height of the image (not required but important)

HTML Images…
> Image Attributes

- **align** Specifies the alignment of an image according to surrounding element. Make an image float (possible values: left, right, middle, top, bottom) – bottom is the default. Obsolete in HTML 5

- **background** Add a background image to an HTML page (not an image attribute)

HTML Images…
> Example (Floating/Aligning)

- **Target**: Display the picture on the left side and the text on the right side.

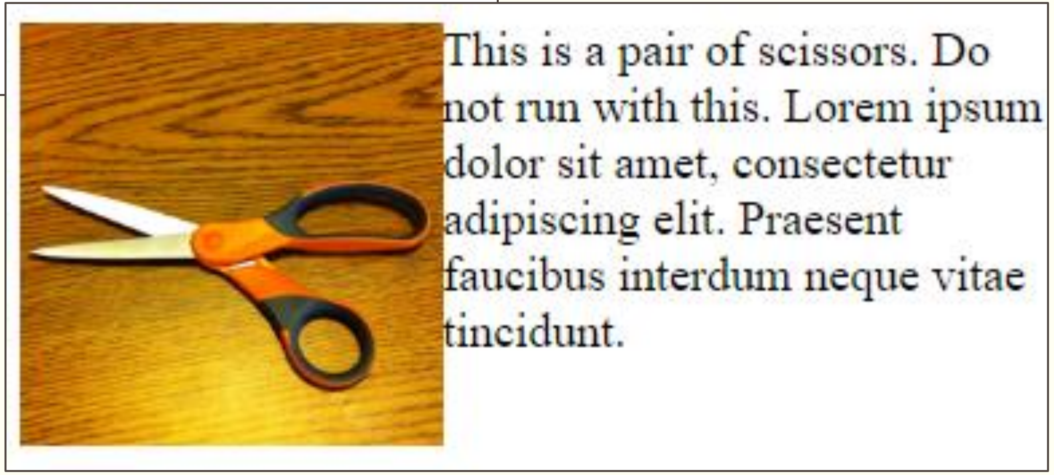
```
<img src="images/scissors.png" width="140" height="140"
    alt="Picture of scissors" title="Running with these is not recommended." />
<p>
    This is a pair of scissors. Do not run with this.
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent
    faucibus interdum neque vitae tincidunt.
</p>
```

This is a pair of scissors. Do not run with this. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent faucibus interdum neque vitae tincidunt.

HTML Images…
> Example (Floating/Aligning)

- Use the float property in the CSS.

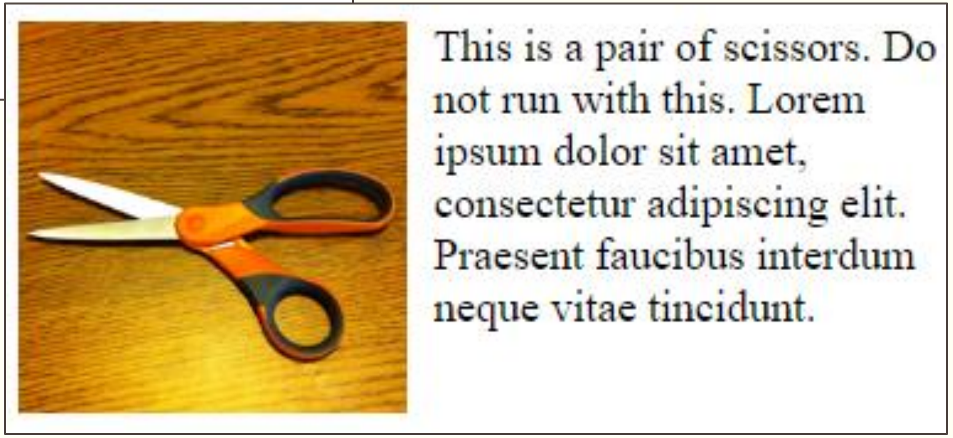
```
<img src="images/scissors.png" width="140" height="140"
    alt="Picture of scissors" title="Running with these is not recommended."
     style="float: left" />
<p>
    This is a pair of scissors. Do not run with this.
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent
    faucibus interdum neque vitae tincidunt.
</p>
```



HTML Images…
> Example (Floating/Aligning)

- The margin property in CSS allows you to create some space between your image and the paragraph.

```html
<img src="images/scissors.png" width="240" height="240"
     alt="Picture of scissors" title="Running with these is not recommended."
     style="float: left; margin-right: 10px; margin-bottom: 5px;
        border: solid black 1px;" />
<p>
    This is a pair of scissors. Do not run with this.
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent
    faucibus interdum neque vitae tincidunt.
</p>
```



HTML Images...
> Break Text

- When text is flowing around an image, sometimes you will need to force some text to start after the image.
- <br clear=“right” /> obsolete in HTML5
- In HTML5: clear property in CSS.
  - clear: left, right, both

This is a pair of scissors. Do not run with this. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent faucibus interdum neque vitae tincidunt.

Ut non ante a eros sodales dapibus a in arcu. Aenean vulputate venenatis dui, eget pellentesque est consectetur id. Curabitur sed magna lacus, et laoreet urna. Vivamus ut adipiscing turpis.

```
<img src="images/scissors.png" width="240" height="240"
     alt="Picture of scissors" title="Running with these is not recommended."
     style="float: right; margin-left: 10px; margin-bottom: 5px;
        border: solid black 1px; padding: 2px;" />
<p>
    This is a pair of scissors. Do not run with this.
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent
    faucibus interdum neque vitae tincidunt.
</p>
<p>
    Ut non ante a eros sodales
    dapibus a in arcu. Aenean vulputate venenatis dui, eget pellentesque
    est consectetur id. Curabitur sed magna lacus, et laoreet urna.
    Vivamus ut adipiscing turpis.
</p>
```

HTML Images…
> Break Text…

```
<img src="images/scissors.png" width="240" height="240"
     alt="Picture of scissors" title="Running with these is not recommended."
     style="float: right; margin-left: 10px; margin-bottom: 5px;
        border: solid black 1px; padding: 2px;" />
<p>
    This is a pair of scissors. Do not run with this.
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent
    faucibus interdum neque vitae tincidunt.
</p>
<p style="clear: right">
    Ut non ante a eros sodales
    dapibus a in arcu. Aenean vulputate venenatis dui, eget pellentesque
    est consectetur id. Curabitur sed magna lacus, et laoreet urna.
    Vivamus ut adipiscing turpis.
</p>
```

HTML Images…
> Break Text…

This is a pair of scissors. Do not run with this. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent faucibus interdum neque vitae tincidunt.

Ut non ante a eros sodales dapibus a in arcu. Aenean vulputate venenatis dui, eget pellentesque est consectetur id. Curabitur sed magna lacus, et laoreet urna. Vivamus ut adipiscing turpis.

Quiz6.html

Using image and float

Lorem ipsum dolor, sit amet consectetur adipisicing elit. Enim corrupti quisquam iure veniam sunt vitae reiciendis mod assumenda dolores explicabo vel veritatis. Ab, quam.

Image Map

- HTML provides a feature to define regions of an image as links to different URLs, this is called Image Map.
- This can be achieved by the **usemap** attribute of an image tag in HTML together with HTML map tag (<map>…</map>.
  - Example: usemap="#scissorsmap"
  - Notice that the value of the usemap is preceded by a # sign. (Fragment Address)
- The map tag has a name (specified by the **name** attribute) that should match the name specified in the **usemap** attribute of the image.
- The map tag has area elements inside of it, to define areas in the image.

Image Map…
> area element attributes

- **shape:** rect, circle or poly
- **coords:** coordinates
  - **rect:** define two points (the upper left and lower right points in x,y format)
    - Example: 0, 0, 50, 50
  - **circle:** define the coordinates of the center of the circle (x, y format) and radius of the circle (r).
    - Example: 150, 25, 25
  - **poly:** define sequence of coordinates (x,y format)

Image Map…
> area element attributes

- **alt:** used when the browser not rendering the map properly
- **title:** displayed when you hover over the map, or used in a descriptive context
- **href:** is a link to a resource

Image Map
> Example

```
<img src="images/scissors-imagemap.png" width="240" height="240"
     alt="Picture of scissors" TITLE="Running with these is not recommended."
     usemap="#scissorsmap" />
<map name="scissorsmap">
    <area shape="rect" coords="0, 0, 50, 50" alt="rect" title="rectangular area" href="rectangle.htm" />
    <area shape="circle" coords="150, 25, 25" alt="circle" title="circular area" href="circle.htm" />
    <area shape="poly"
          coords="13, 93, 101, 114, 148, 106, 193, 88, 233, 111, 220, 129,
    140, 134, 191, 163, 203, 187, 193, 202, 173, 208, 148, 181, 99,
    137, 5, 130, 68, 120"
          alt="poly" title="poly area for scissors" href="scissors.htm" />
</map>
```

HTML Links/ Hyperlinks/ Hypertext links

- Imagine a book.
  - Go to table of content.
  - Turn the page until we find the page
- **A hyperlink is a reference (an address) to a resource on the web.**
- Hyperlinks can point to any resource on the web: an HTML page, an image, a sound file, a movie, etc.
- The HTML anchor element <a>, is used to define hyperlinks on a document.
- Example:

```
<a href="http://www.w3schools.com">This is a link</a>
```

Note: The link (URL) is specified in the href attribute.

HTML Links/ Hyperlinks/ Hypertext links
> Link Colors

- By default, links will appear as follows in all browsers:
    - An unvisited link is **underlined and blue**
    - A visited link is **underlined and purple**
    - An active link is **underlined and red**

HTML Links/ Hyperlinks/ Hypertext links > The target Attribute

- The target attribute specifies where to open the linked document.
- The example below will open the linked document in a new browser window or a new tab
  - **Example:**

    Visit W3Schools!
- target possible values:
  - **_blank** - Opens the linked document in a new window or tab
  - **_self** - Opens the linked document in the same window/tab as it was clicked (this is default)

URL

- Stands for Uniform Resource Locator
  - Way of specifying the location of resource on the internet
  - Resources can be
    - Webpage
    - Image
    - File
    - Any kind of service…

HTML Links/ Hyperlinks/ Hypertext links > URLs

- URL: stands for Uniform Resource Locator
  - It is a way of specifying a location of a resource on the internet.
  - A resource can be a webpage, an image, a file of some kind, or really any kind of service. It doesn't even always map to a particular object on a particular server, It is really just an address.

Protocol / Scheme

Port

FileName

Fragment Identifier

http://www.HiLCoE.net:80/webdevanddesign/index.htm?query#location

Query String

Host / Domain name

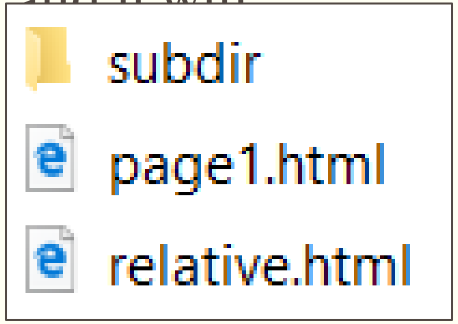
Path

HTML Links/ Hyperlinks/ Hypertext links > URLs: Example

- http://www.google.com/
  - Specifies a hostname (www.google.com) and a path (/)
- http://www.google.com/contact/
  - Specifies a hostname and a path (/contact/)
  - Refers to the default resource on the directory
  - /contact/ is called a relative URL, and specifies the contact directory on the current host.
  - If you put a relative URL (/contact/) on a link on a document the browser will complete it to include the host name of the host for the current document.
- Another relative url: filename.html
  - **The web browser will complete it with the host name and directory of the current document.**

HTML Links/ Hyperlinks/ Hypertext links > Relative URL

- **Relative URLs are URLs that don’t specify a complete hosting path**.
- The browser, when it finds a relative URL, it will construct a complete URL and it will use it as the absolute URL of the resource.




```html
<p>
    Here is a link to <a href="page1.html">page 1</a>.
    The text around the link is not part of the link.
</p>
```

HTML Links/ Hyperlinks/ Hypertext links > Fragments (Linking within a page)

- In HTML it is possible to link to a part of a page causing the browser to scroll to that point. This is achieved through fragments.
- Fragments are used for linking to a section of a page.
- **They are useful when we have a large page.**
- How do you achieve that (fragments.html)
  - Create a link to the section you want to be taken to:

```
<div id="top">
    <h1>Countries</h1>
    <p>
        <a href="#AFG">Afghanistan</a> |
        <a href="#ALB">Albania</a> |
        <a href="#DZA">Algeria</a> |
        <a href="#ASM">American Samoa</a> |
        <a href="#AND">Andorra</a> |
```

#AFG is a relative URL. (Example: http://localhost:53331/fragments.html#AFG)

HTML Links/ Hyperlinks/ Hypertext links > Image Links

- Using images as link is a very common thing.
- This is achieved through adding an <img> tag inside an <a> tag.
-
  <img src="smiley.gif" alt="HTML tutorialstyle="width:42px;height:42px;border:0;">

The <div> Tag

- The <div> tag defines a division or a section in an HTML document.
- The <div> tag is used to group block-elements to format them with CSS.
- Block element

```html
<div style="font-size:24px; color:red">DIV example</div>
<p>This one is <span style="color:red; font-weight:bold">only a test</span>.</p>
```

DIV example

This one is **only a test**.

The <span> Tag

- Inline style element
- Useful for modifying a specific portion of text
  - Don't create a separate area (paragraph) in the document

```
<p>This one is <span style="color:red; font-weight:bold">only a test</span>.</p>
<p>This one is another <span style="font-size:32px; font-weight:bold">TEST</span>.</p>
```

This one is **only a test**.

This one is another **TEST**.

Semantic Elements

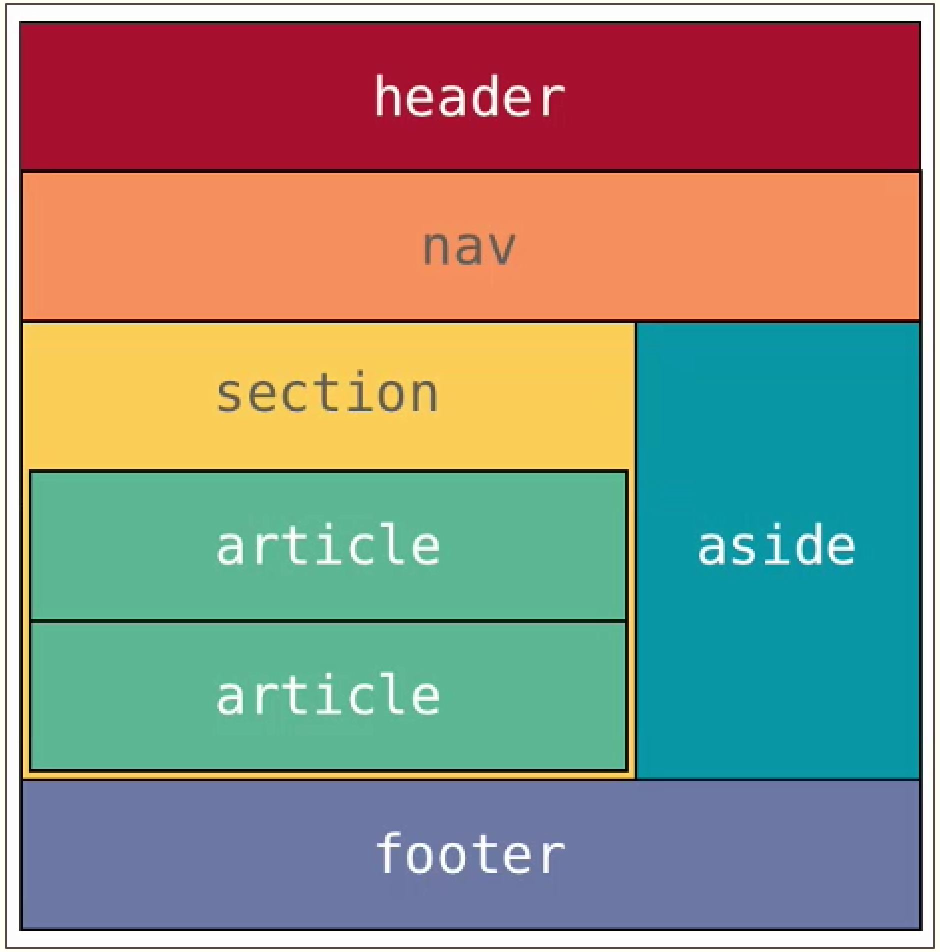
- Semantics is the study of the meanings of words and phrases in language.
- **Semantic elements are elements with a meaning**.
- A semantic element clearly describes its meaning to both the browser and the developer.
- Examples of **non-semantic** elements: <div> and <span> - Tells nothing about its content.
- Examples of **semantic** elements: <form>, <table>, and <img> - Clearly defines its content.

Semantic Elements(https://www.semrush.com/blog/semantic-html5-guide/)

- HTML5 introduces a number of block level sematic elements for organizing the content of a webpage.
- These are block level elements with no inherent presentation properties, just like div.
- It is how they are designed to be used that makes them distinct.
- These elements are designed to **create structure for your documents**, so that it makes sense not only to the reader but to other processes that may need to understand the structure of your document, ***like search engines, archival processors, non-visual browsers for the blind or even future AI processes that we haven't even thought of yet.***
- The point is it should be possible for a relatively **unsophisticated machine to create an outline of your document** so it knows which part are relevant for a particular purpose.

Sematic Elements

- Hypothetical document structure (Example)
  - From the diagram it should be easy to see we have a fairly typical webpage.
    - Has a header
    - Has a footer
    - A section with a couple of articles
    - Aside bar
    - Footer
  - It is easy to see because it all labeled for us.
- **That is the point of semantic markup, you are creating labels that make it obvious what parts of your page serves what purpose.**



machines like Googlebot and Bingbot LOVE this in place of DIV.

<header>

<nav>

<main>

<article>

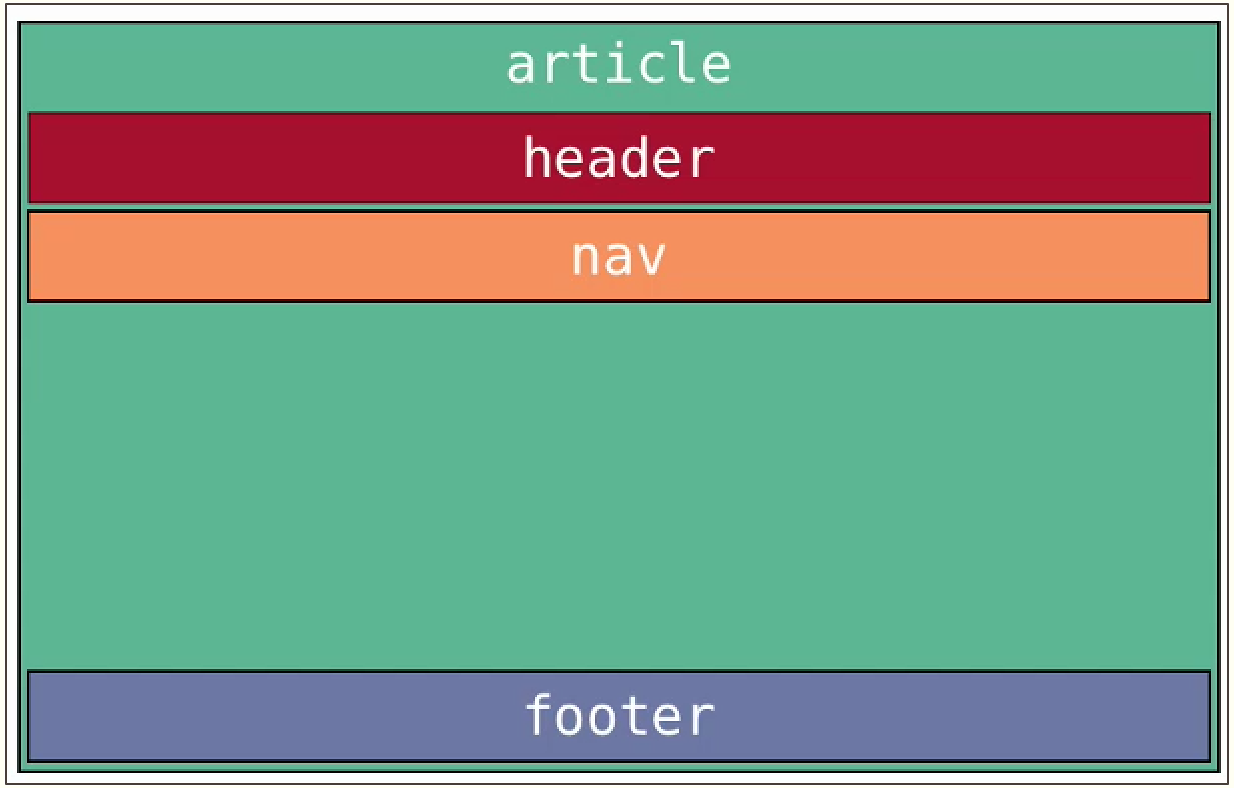
<section>

<aside>

<footer>

Sematic Elements

- Each semantic segments may also have its own semantic segments (nesting). This is common and encouraged.
- Example: If an article has
  - A header
  - A footer
  - Its own navigation.



Document Outlines

- The document **outline is the structure of a document**, generated by the document's headings, form titles, table titles, and any other appropriate landmarks to map out the document.
- The user agent can apply this information **to generate a table of contents**, for example. This table of contents could then be used by assistive technology to help the user, or be parsed by a machine like a search engine to improve search results.

Document Outlines

- Major sectioning elements (This are a list of sectioning elements that are intended to be used where you would otherwise use a <div>).
  - **section**: represents a generic section of the document. It is not a generic container for styling purposes like <div>, rather it is intended to indicate a semantic section of the document that is not otherwise covered by another sectioning element.
  - **article**: represents a self-contained composition within a document. The HTML5 specification says this could be, a form post, a magazine or a newspaper article, a blog entry, a user submitted comment, an interactive widget or gadget or any other independent item of content.
  - **nav:** represents the section of the page that contains a set of links to other pages or other resources. (You might not use them all the time)
  - **aside:** represents a part of the page that is tangentially related to the content around it like a sidebar or other semi-related contents.

Document Outlines

- The following sematic elements are not sections but are used with in sections. This elements are designed to **describe parts of a section for the benefit of document outlines**, summaries and other tools.
  - **header:** is for a header part of a section. It represents a group of introductory or navigational aids.
  - **footer:** contains information about its section like author information, links to related content, copyright etc. It is typically at the end of a section, but it doesn’t have to be.
  - **hgroup:** used to group heading tags together. When a number of <h> tags grouped into an <hgroup> element the first example of the highest ranking heading will be used as the text for the entire group.

Document Outlines
> Example: Application of the Algorithm

```
<nav id="top-menu">
    ...
</nav>

<header>
    <hgroup>
        <h1> Scissors </h1>
        ...
    </hgroup>
</header>

<article class="story">
    <h1> The Story of Scissors </h1>
    ...
</article>

<aside class="about">
    <header>
        <h1> About These Scissors </h1>
        ...
    </header>
    <section>
        <hgroup>
            <h1> Blades </h1>
            ...
        </hgroup>
            ...
    </section>
    <section>
        <hgroup>
            <h1> Handles </h1>
            ...
        </hgroup>
            ...
    </section>
    <section>
        <hgroup>
            <h1> Color </h1>
            ...
        </hgroup>
        ...
    </section>
</aside>
```

```
1. Scissors
  1.1. untitled section nav
  1.2. The Story of Scissors
  1.3. About These Scissors
    1.3.1. Blades
    1.3.2. Handles
    1.3.3. Color
```

Quiz 7…show the structure of this page using html5 semantic elements

SERVICES
ORGANIZATION
PROJECTS
PUBLICATIONS
JOBS
FOR STAFF
HELP
SEARCH

STANFORD UNIVERSITY

Information Technology
Systems and Services

ITSS provides services to the research, instructional, and administrative communities at Stanford University. These services include the planning, development, acquisition, and operation of institutional networking and telecommunications services, information systems, data administration, and information technology infrastructure support.

Within these pages you will find detailed descriptions of the services, equipment and groups that provide the above support.

Throughout the year, ITSS introduces new computing services and enhances existing ones. **"Changes to Computing Services"** is a quarterly publication that summarizes those upcoming changes, including highlighting changes that **require** that you, or your local computer support staff, take some action.

New! ITSS Services Monitoring

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Comments?

Last modified: May 11, 2000

Investors' Day Award

The Winner of the 2002 Investors' Day Award in the area of Information Communication Technology is HiLCoE School of Computer Science and Technology.

HiLCoE 2018 Graduation at UNCC

News

- New HiLCoE Campus
- First MSc batch graduation
- New Students Registration in progress
- Graduates of 2014

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Links to Research Resources

- Open Access Library
- ACM Digital Library
- IEEE Digital Library
- ScienceDirect
- Springer Link
- AIS Link
- Google Scholar
- AAU Research Repository

Links to Other Useful Resources

- Free Tutorials

HiLCoE Brochure 2018

- Hilcoe Brochure 2018

Figure Element

- The figure element is for representing figures in a document.
- You may use the figure element for illustrations, diagrams, photos or any other content that is illustrative in nature and perhaps uses a caption.
- The <figcaption> element inside the figure tag allows you in specifying a caption for the figure.
    - The caption can be displayed at the top of the figure using the caption-side property inside the style attribute and setting its value to **top**.

Figure Element
> Example

```
<figure>
    <img src="images/paper.png" alt="a rockin'piece of paper!" title="Paper beats rock!" />
    <figcaption> Paper always beats Rock. </figcaption>
</figure>
```

Paper always beats Rock.

Collapsible Details

- HTML5 introduces a new feature for collapsible content called details.
- Summary can be specified using the summary element inside the details tag.

```
<details>
    <summary>Copyright 1999-2014.</summary>
    <p> - by Contoso Plc. All Rights Reserved.</p>
    <p>All content and graphics on this web site are the property of the company Contoso Plc.</p>
</details>
```

▶ Copyright 1999-2014.

▼ Copyright 1999-2014.

- by Contoso Plc. All Rights Reserved.

All content and graphics on this web site are the property of the company Contoso Plc.

Embedding Audio

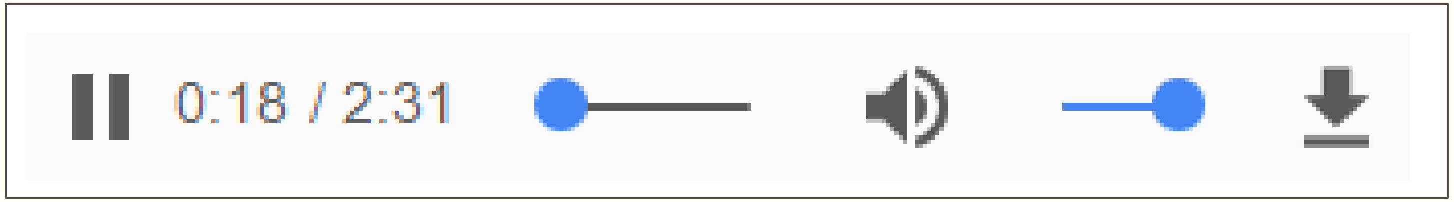
- HTML5 provides a simple and powerful feature for embedding audio files in your webpages.
- This is achieved through the <audio> tag.
- The audio tag contains a child element called source that allows you in specifying audio sources to our audio tag.
  - Multiple sources can be specified so that the player supports different kinds of audio encoding. The reason for specifying multiple sources for a single media is that different browsers supports different kind of audio encoding. The browser will look into the source and loads the supported audio encoding.
- If HTML5 audio is not supported on a given browser, one can also specify a text that is going to be displayed when the browser fails to load the audio.

Embedding Audio

- The **controls** attribute on the audio tag allows us in enabling controls for our audio player. Failing to specify the attribute results in a non visible player which one can implement a custom buttons for controlling the player.
- The **autoplay** attribute specifies that the audio should be played automatically once the page is loaded.
- The **loop** attribute enables your audio to be played again and again.
- The **src** attribute on the source element specifies the URL for the audio where as the **type** attribute specifies the encoding.

Embedding Audio
> Example

```
<audio id="audio1" controls autoplay loop>
    <source src="media/song.m4a" type="audio/x-aac" />
    <source src="media/song.mp3" type="audio/mpeg" />
    <source src="media/song.ogg" type="audio/ogg" />
    <p> Your browser does not support the HTML5 audio feature. </p>
</audio>
```



Embedding Video

- HTML5 provides a simple and powerful feature for embedding video in your webpages.
- This is achieved through the <video> tag.
- The video tag contains a child element called source that allows you in specifying video sources to our video tag.
  - Multiple sources can be specified so that the player supports different kinds of video encoding. The reason for specifying multiple sources for a single media is that different browsers supports different kind of video encoding. The browser will look into the source and loads the supported video encoding.
- If HTML5 video is not supported on a given browser, one can also specify a text that is going to be displayed when the browser fails to load the video.

Embedding Video

- The **controls** attribute on the video tag allows us in enabling controls for our video player. Failing to specify the attribute results in a video player with no controls. one can implement a custom buttons for controlling the player.
- The **autoplay** attribute specifies that the video should be played automatically once the page is loaded.
- The **loop** attribute enables your video to be played again and again.
- The **width** and **height** allows you in scaling your video player.
- The **src** attribute on the source element specifies the URL for the video where as the **type** attribute specifies the encoding.

Embedding Video
> Example

```html
<video id="video1" width="860" height="480" controls autoplay loop>
    <source src="media/video.m4v" type="video/mp4" />
    <source src="media/video.webm" type="video/webm" />
    <source src="media/video.ogg" type="video/ogg" />
    <p> Your browser does not support the HTML5 video feature. </p>
</video>
```

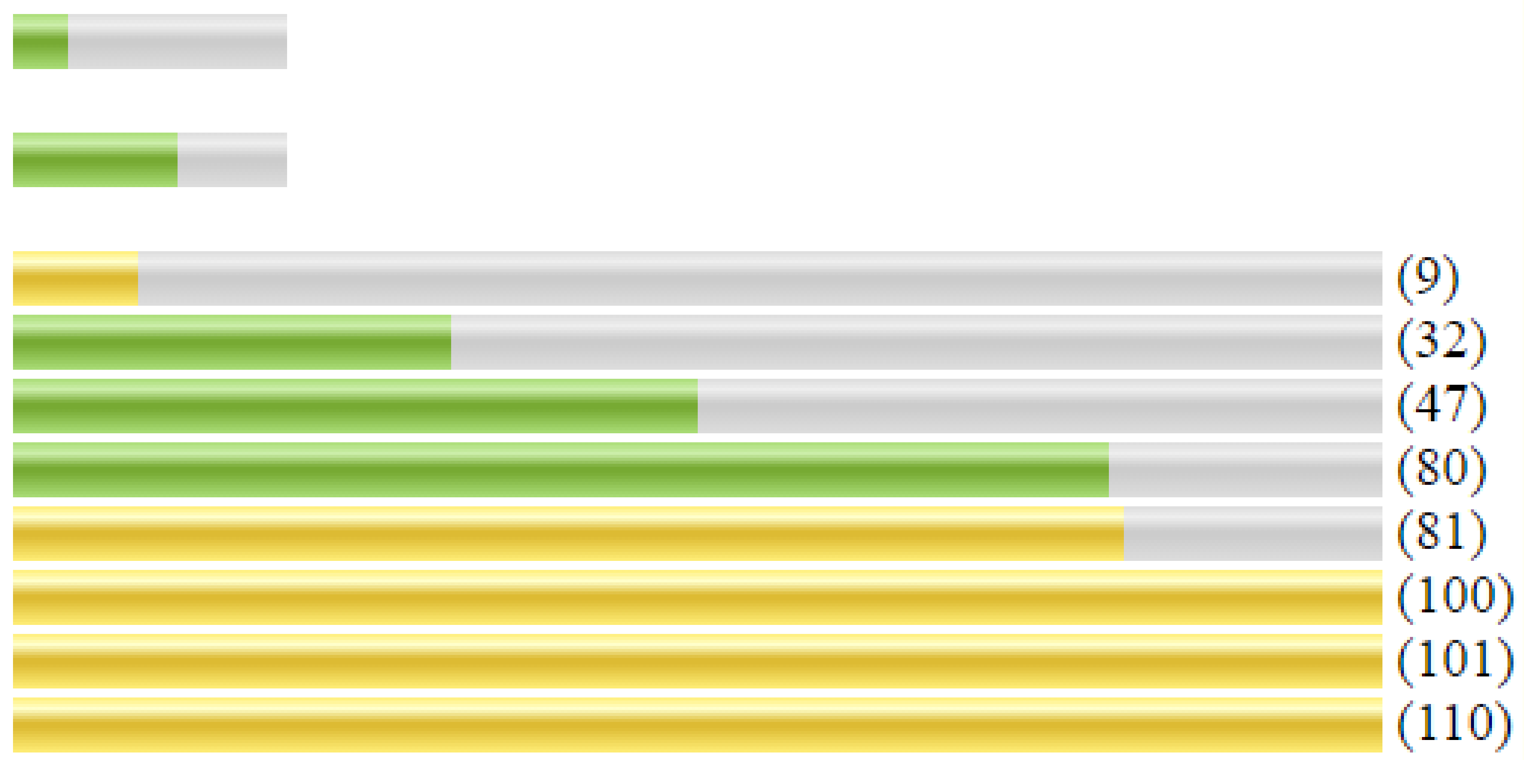
The Meter Element
> Displaying relative values (static)

- HTML5 provides an element for **displaying static value** in a range.
- This element is called the meter element.
- Attributes:
  - value: the current value on the meter
  - min: the minimum value on the meter
  - max: the maximum value on the meter
  - low: specifies below what value is considered low
  - high: specifies above what value is considered high
  - optimum: specifies what value is considered optimum

The Meter Element
> Example

```
<p>
    <meter value="2" min="0" max="10">2 out of 10</meter>
</p>
<p>
    <meter value="0.6">60%</meter>
</p>
<p>
    <meter value="9" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">9 of 100</meter> (9)<br />
    <meter value="32" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">32 of 100</meter> (32)<br />
    <meter value="50" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">47 of 100</meter> (47)<br />
    <meter value="80" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">80 of 100</meter> (80)<br />
    <meter value="81" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">81 of 100</meter> (81)<br />
    <meter value="100" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">100 of 100</meter> (100)<br />
    <meter value="101" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">101 of 100</meter> (101)<br />
    <meter value="110" min="0" max="100" low="10" high="80" optimum="47" style="width:400px">110 of 100</meter> (110)<br />
</p>
```

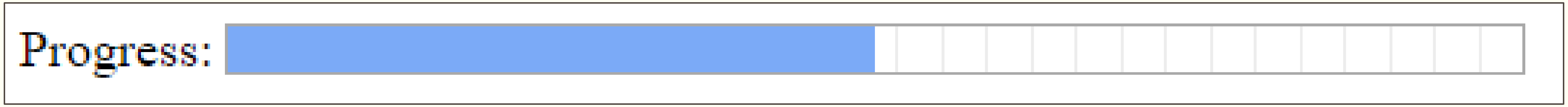
The Meter Element
> Example: Result



The Progress Element

- The progress element is recommended for displaying **a progress bar**. (Dynamic Data)
- The **value** attribute allows in specifying the current value on the progress bar.
- The **max** attribute allows in specifying the maximum value on the progress bar.

```
<p>
    Progress: <progress id="p1" value="50" max="100" style="width: 400px"><span id="s1">50</span>%</progress>
</p>
```



HTML Table

- Tables are a very useful feature in HTML.
- Tables are defined with the <table> tag.
- The **caption** tag allows you to set a title for your table, it should be specified before any rows of data inside the table element.
- A table is divided into rows (with the <**tr**> tag – table row).
- A table row contains cells/data cells. There are two types of cells
  - Header Cells (defined by the <th> tag)
  - Data Cells (defined by the <td> tag)
- A <td> and <th> tag can contain text, links, images, lists, forms, other tables, etc.

HTML Table...

Some Favorite Albums

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Birds of Fire	Mahavishnu Orchestra	The ultimate fusion album.

```
<table>
    <caption>
        Some Favorite Albums
    </caption>
    <tr>
        <th> Title </th>
        <th> Artist </th>
        <th> Comment </th>
    </tr>
    <tr>
        <td> Electric Ladyland </td>
        <td> Jimi Hendrix </td>
        <td> Revolutionary </td>
    </tr>
    <tr>
        <td> Revolver </td>
        <td> The Beatles </td>
        <td> Their best work. </td>
    </tr>
    <tr>
        <td> Apostrophe </td>
        <td> Frank Zappa </td>
        <td> Brilliant mix of rock and jazz. </td>
    </tr>
    <tr>
        <td> Birds of Fire </td>
        <td> Mahavishnu Orchestra </td>
        <td> The ultimate fusion album. </td>
    </tr>
</table>
```

HTML Table…Border

- The default layout for a table is borderless (not recognizable in its native state).
  - Set the border property in the style attribute for table, td and th.
  - style="border: 1px solid black;"

Some Favorite Albums

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Birds of Fire	Mahavishnu Orchestra	The ultimate fusion album.

HTML Table…

- By default, each of table elements (cells) has its own border rather than sharing borders with the elements in the next room.
- To avoid that one can use the border-collapse property in the table's style attribute and set it to collapse.
  - style="border-collapse: collapse" (separate is the default)

Some Favorite Albums

Title	**Artist**	**Comment**
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Birds of Fire	Mahavishnu Orchestra	The ultimate fusion album.

Table HTML…

- Caption of a table can be set to be displayed at the top or bottom of a page. This can be achieved using the caption-side property inside the style attribute of the caption tag and set it to bottom;
  - style="caption-side: bottom" (top is the default)

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Birds of Fire	Mahavishnu Orchestra	The ultimate fusion album.

Some Favorite Albums

Table HTML…

- Before HTML5, spacing between a border of a cell and its content can be modified by using cellpadding, and the spacing between neighboring cell borders can be modified using cellspacing. These properties are now obsolete.
- In HTML5, spacing can be achieved by using the padding property inside the style attribute of the <td> or <table> element and set it to some value in pixel.
  - Example: style="padding: 5px;" (top is the default)

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Birds of Fire	Mahavishnu Orchestra	The ultimate fusion album.

Some Favorite Albums

HTML Table…

- The spacing between cells can be modified by using the border-spacing property on the table’s style attribute which is set in pixels too.
  - For this to work the border-collapse property of the style attribute of the table should be set to separate.
  - Example: style=“border-collapse: separate; border-spacing: 20px;”

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Birds of Fire	Mahavishnu Orchestra	The ultimate fusion album.

Some Favorite Albums

HTML Table…

- Table cells that span more than one row or more than one column can be achieved through the rowspan and colspan attributes respectively.

```html
<table style="border: 1px solid black;">
    <tr>
        <th style="border: 1px solid black;">Name</th>
        <th style="border: 1px solid black;" colspan="2">Telephone</th>
    </tr>
    <tr>
        <td style="border: 1px solid black;">Ezra Tadios</td>
        <td style="border: 1px solid black;">+251 (911) 37-54-87</td>
        <td style="border: 1px solid black;">+251 (114) 27-52-88</td>
    </tr>
</table>
```

Name	Telephone	
Ezra Tadios	+251 (911) 37-54-87	+251 (114) 27-5

HTML Table...

```
<table style="border: 1px solid black;">
    <tr>
        <th style="border: 1px solid black;">Name</th>
        <td style="border: 1px solid black;">Ezra Tadios</td>
    </tr>
    <tr>
        <th rowspan="2" style="border: 1px solid black;">Telephone</th>
        <td style="border: 1px solid black;">+251 (911) 37-54-87</td>
    </tr>
    <tr>
        <td style="border: 1px solid black;">+251 (114) 27-52-88</td>
    </tr>
</table>
```

<table>
<tr><th>Name</th><td>Ezra Tadios</td></tr>
<tr><th rowspan="2">Telephone</th><td>+251 (911) 37-54-87</td></tr>
<tr><td>+251 (114) 27-52-88</td></tr>
</table>

HTML Table…

- Using the <colgroup> and <col> elements you can apply styles to columns and group of columns in your tables.
- The <colgroup> element must always go after the caption element if there are any and before any <thead>, <tbody>, <tfoot> or <tr> elements.

HTML Table...

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Footer C1	Footer C2	Footer C3

Some Favorite Albums

```
<table>
    <caption style="caption-side: bottom">
        Some Favorite Albums
    </caption>
        <colgroup>
            <col style="background-color: #fcd" />
            <col style="background-color: #cfd" />
            <col style="background-color: #cdf" />
        </colgroup>
    <thead>
        <tr>
            <th> Title </th>
            <th> Artist </th>
            <th> Comment </th>
        </tr>
    </thead>
    <tfoot>
        <tr>
            <td> Footer C1 </td>
            <td> Footer C2 </td>
            <td> Footer C3 </td>
        </tr>
    </tfoot>
```

```
    <tbody>
        <tr>
            <td> Electric Ladyland </td>
            <td> Jimi Hendrix </td>
            <td> Revolutionary </td>
        </tr>
        <tr>
            <td> Revolver </td>
            <td> The Beatles </td>
            <td> Their best work. </td>
        </tr>
        <tr>
            <td> Apostrophe </td>
            <td> Frank Zappa </td>
            <td> Brilliant mix of rock and jazz. </td>
        </tr>
    </tbody>
</table>
```

HTML Table...

- The **span** attribute on the <col> element allows you to represent more than one adjacent columns.

```
<table>
    <caption style="caption-side: bottom">
        Some Favorite Albums
    </caption>
        <colgroup>
            <col span="2" style="background-color: #fcd" />
            <col style="background-color: #cfd" />
        </colgroup>
    <thead>
        <tr>
            <th> Title </th>
            <th> Artist </th>
            <th> Comment </th>
        </tr>
    </thead>
    <tfoot>
        <tr>
            <td> Footer C1 </td>
            <td> Footer C2 </td>
            <td> Footer C3 </td>
        </tr>
    </tfoot>
```

```
    <tbody>
        <tr>
            <td> Electric Ladyland </td>
            <td> Jimi Hendrix </td>
            <td> Revolutionary </td>
        </tr>
        <tr>
            <td> Revolver </td>
            <td> The Beatles </td>
            <td> Their best work. </td>
        </tr>
        <tr>
            <td> Apostrophe </td>
            <td> Frank Zappa </td>
            <td> Brilliant mix of rock and jazz. </td>
        </tr>
    </tbody>
</table>
```

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Apostrophe	Frank Zappa	Brilliant mix of rock and jazz.
Footer C1	Footer C2	Footer C3

Some Favorite Albums

HTML Table
> Semantic Parts

- For most part, semantic table tags have no presentation meaning, except the <tfoot> element always end up at the bottom of the table. They are sectioning elements.
  - **<thead>**: The thead element should be the first element in a table after a caption if there is a caption.
  - **<tfoot>**: can be at the top of the body or the bottom. Have a presentation meaning, that it always end up at the bottom.
  - **<tbody>**: can be anywhere after the <thead>

HTML Table…
> Semantic Parts…

Title	Artist	Comment
Electric Ladyland	Jimi Hendrix	Revolutionary
Revolver	The Beatles	Their best work.
Footer C1	Footer C2	Footer C3

Some Favorite Albums

```html
<table>
    <caption style="caption-side: bottom">
        Some Favorite Albums
    </caption>
    <thead>
        <tr>
            <th> Title </th>
            <th> Artist </th>
            <th> Comment </th>
        </tr>
    </thead>
    <tfoot>
        <tr>
            <td> Footer C1 </td>
            <td> Footer C2 </td>
            <td> Footer C3 </td>
        </tr>
    </tfoot>
    <tbody>
        <tr>
            <td> Electric Ladyland </td>
            <td> Jimi Hendrix </td>
            <td> Revolutionary </td>
        </tr>
        <tr>
            <td> Revolver </td>
            <td> The Beatles </td>
            <td> Their best work. </td>
        </tr>
    </tbody>
</table>
```

Quiz 9:-Use HTMl5 sematic table elements<thead><tbody><tfoot> to create the ff

Example of Colgroup Tag

Sr.No	Product	Price
1	Rice	85
2	Butter	260
3	Mango	125

HTML Frames

- HTML frames helps in displaying more than one HTML document in the same browser window.
- Each HTML document is called a frame, and each frame is independent of the other.
- There are two distinct versions of HTML frames.
  - The legacy HTML Frame (Frameset) (HTML3 and HTML4, now obsolete)
  - The iFrame (supported in HTML5)

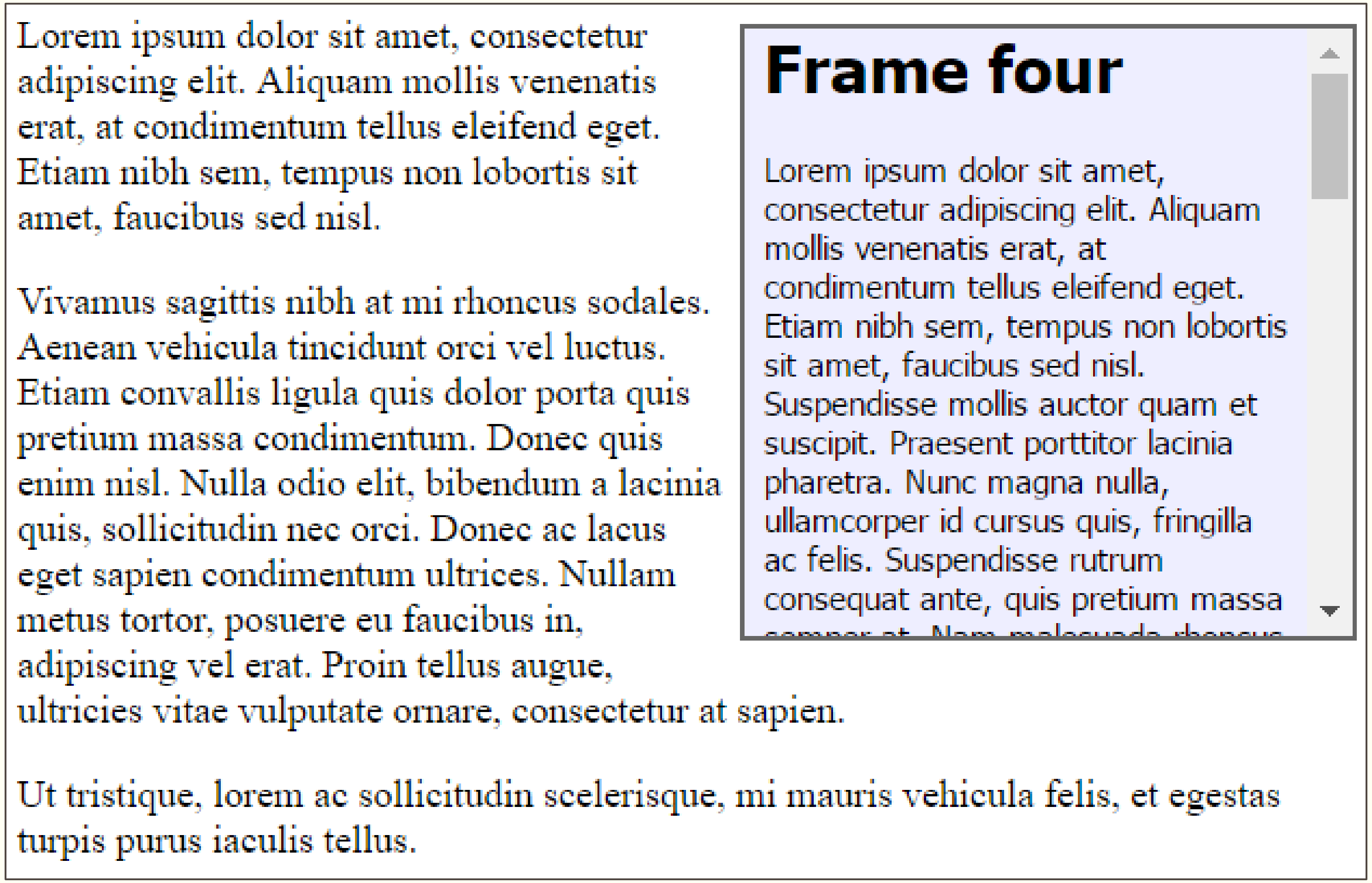
HTML Frames…
> iframe

- The modern version of frames.
- More flexible than frameset, and can be styled using CSS.
- It is recommended you use iFrame for any frames applications you may have now and in the future.
- The **src** attribute on iframe allows you in specifying a URL for a resource that you want to open in the frame.

HTML Frames…
> iframe

```
<iframe id="frame1" style="display: block; float:right; width: 250px;
    height:250px; margin: 4px 0 4px 4px;
    padding: 0; border: 2px solid #666" src="frames/f4.html"></iframe>
<p>
    Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam
    mollis venenatis erat, at condimentum tellus eleifend eget. Etiam
    nibh sem, tempus non lobortis sit amet, faucibus sed nisl.
</p>
<p>
    Vivamus sagittis nibh at mi rhoncus sodales. Aenean vehicula
    tincidunt orci vel luctus. Etiam convallis ligula quis dolor porta
    quis pretium massa condimentum. Donec quis enim nisl. Nulla odio
    elit, bibendum a lacinia quis, sollicitudin nec orci. Donec ac lacus
    eget sapien condimentum ultrices. Nullam metus tortor, posuere eu
    faucibus in, adipiscing vel erat. Proin tellus augue, ultricies
    vitae vulputate ornare, consectetur at sapien.
</p>
<p>
    Ut tristique, lorem ac sollicitudin scelerisque, mi mauris vehicula
    felis, et egestas turpis purus iaculis tellus.
</p>
```

HTML Frames…
> iframe




IFrame Sample Website
Profile
Videos
Photos
Web Images Maps News Orkut Groups Gmail more ▾
iGoogle | Search settir
Google
India
Advanced Search
Language Tools
Google Search
I'm Feeling Lucky
Search: the web pages from India
Google.co.in offered in: Hindi Bengali Telugu Marathi Tamil Gujarati Kannada Malayalam Punjabi
Advertising Programs - About Google - Go to Google.com
©2009 - Privacy


Registration Form - Mozilla Fi...
File Edit View History Bookmarks Tools Help
http:/
Registration Form
User name:
Password:
Gender: Male Female
Click to accept our terms:
Register Reset
Done
HTML

HTML FORMS


First Name:
Last Name:
Age:
1-17 yrs
18 yrs and over
I would like to learn to work with:
watercolors
acrylics
pastels
I am interested in art lessons because:
Send me an application now!

HTML Form

- The form interface is extremely important in HTML. It is how HTML does interactivity.
- A form is an area that can contain form elements.
- HTML forms are used to select different kinds of user input.
- Form elements are elements that allow the user to enter information in a form. Like text fields, textarea fields, drop-down menus, radio buttons and checkboxes and submit button, etc.
- A form is defined with the **<form>** tag.

HTML Form…

- The general syntax:

```
<form>
    …
    input elements
    …
</form>
```

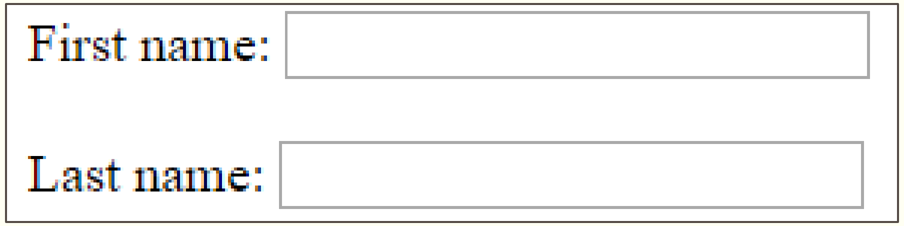
HTML Input

- The most important form element is the <input> element.
- The <input> element is used to accept user information.
- An <input> element can vary in many ways, depending on the type attribute. An <input> element can be of type text field, checkbox, password, radio button, submit button and more.
- The default type for an HTML input element is text.
- The value attribute for an HTML form control sets the value of the HTML control.

HTML Text Elements
> Text: <input type=“text” />

- Text fields are used when you want the user to type letters, numbers, etc. in a form.
- <input type=“text”> defines a one-line input field that a user can enter text into.
- Example:

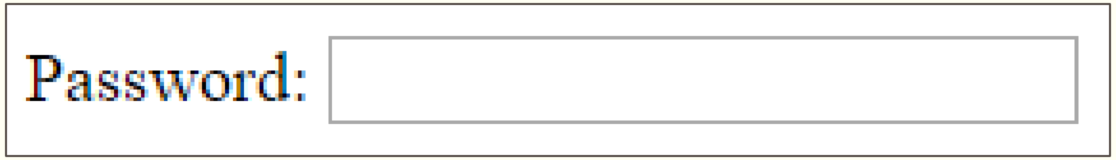
```
First name: <input type="text" name="firstname" /> <br /> <br />
Last name: <input type="text" name="lastname" /> <br />
```



HTML Text Elements
> Password: <input type="password" />

- Password fields are used when you want the user to type an obscured text (like password) in a form.
- <input type="password" /> defines a one-line password field that a user can enter text into.
- Example:

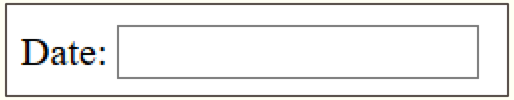
```
<p> Password: <input type="password" name="password1" /> </p>
```



HTML Text Elements
> Date: <input type="date" />

- Date fields are used when you want the user to type a date in a form.
- <input type="date" /> defines a one-line date field that a user can enter text into.
- Date fields can have an automatic date picker on some browsers like Google Chrome.
- Example:
- **<label for="start-date">Start Date:</label><input type="date" id="start-date" name="start-date" min="2024-01-01" max="2024-12-31">**

```
<p>
    Date: <input type="date" name="date1"
             pattern="\d{4}-\d{2}-\d{2}" title="YYYY-MM-DD" />
</p>
```



HTML Text Elements
> Date: <input type="tel" />

- specifies that the input field is intended for telephone numbers.

pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}" defines a pattern using a regular expression:

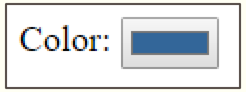
placeholder="###-###-####" provides a hint to the user about the expected format

```
<input type="tel" pattern="[0-9]{3}-[0-9]{3}-[0-9]{4}" placeholder="###-###-####">
```

HTML Text Elements
> Color: <input type="color" />

- Color fields are used when you want the user to select a color in a form.
- <input type="color" /> defines a one-line color field that a user can select a color into.
- The color field, when clicked, provides the operating systems color picker to choose from.
- Example:

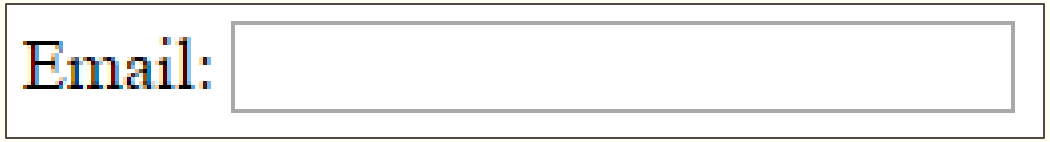
```
<p> Color: <input type="color" name="color1" value="#336699" /> </p>
```



HTML Text Elements
> Email: <input type=“email” />

- Email fields are used when you want the user to input an email in a form.
- <input type=“email” /> defines a one-line email field that a user can enter an email into.
- The **multiple** attribute allows you to specify multiple email addresses separated by a comma.
- Example:

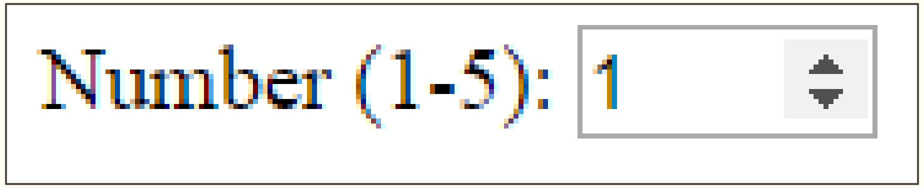
```
<p> Email: <input type="email" name="email1" multiple /> </p>
```



HTML Text Elements
> Number: <input type="number" />

- Number fields are used when you want the user to input a number in a form.
- <input type="number" /> defines a one-line number field that a user can enter a number into.
- The **min** and **max** attributes allows you in specifying the minimum and maximum numbers allowed in the input field respectively.

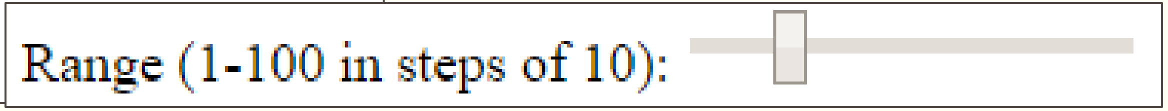
```
<p> Number (1-5): <input type="number" name="number1" min="1" max="5" value="1" /> </p>
```



HTML Text Elements
> Range: <input type=“range” />

- Range fields are used when you want the user to select a value in a given specific range in a form.
- <input type=“range” /> defines a one-line graphical range selector that a user can drag to select a value from it.
- The **min** and **max** attributes allows you in specifying the minimum and maximum values for the range field respectively.
- The **step** attribute allows you in specifying by how much the step indicator can be moved in the provided range. (The default is 1)

```
<p>
    Range (1-100 in steps of 10):
    <input type="range" name="range1" min="0" max="100"
           step="10" value="20" />
</p>
```



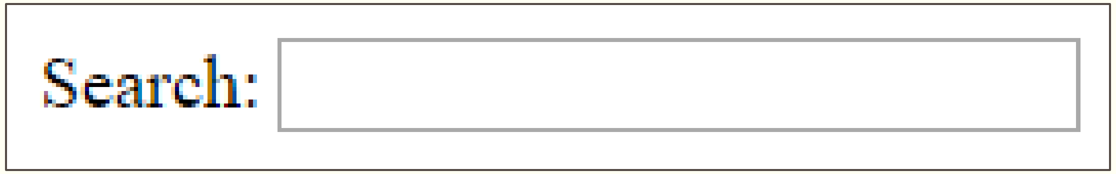
HTML Text Elements
> Search: <input type=“search” />

- Search fields are used when you want the user to type letters, numbers, etc. in a form just like a text field. The only difference with text field is that it is formatted to look like a search box.

- <input type=“search”> defines a one-line input field that a user can enter a search text into.

- Example:

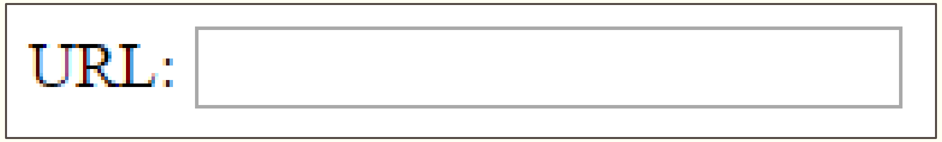
```
<p> Search: <input type="search" name="search1" /> </p>
```



HTML Text Elements
> URL: <input type=“url” />

- URL fields are used when you want the user to enter a valid URL in a text box. This is just like a text field with the exception that the URL field is checked if the users input is a valid URL.
- <input type=“url”> defines a one-line input field that a user can enter a url into.
- Example:

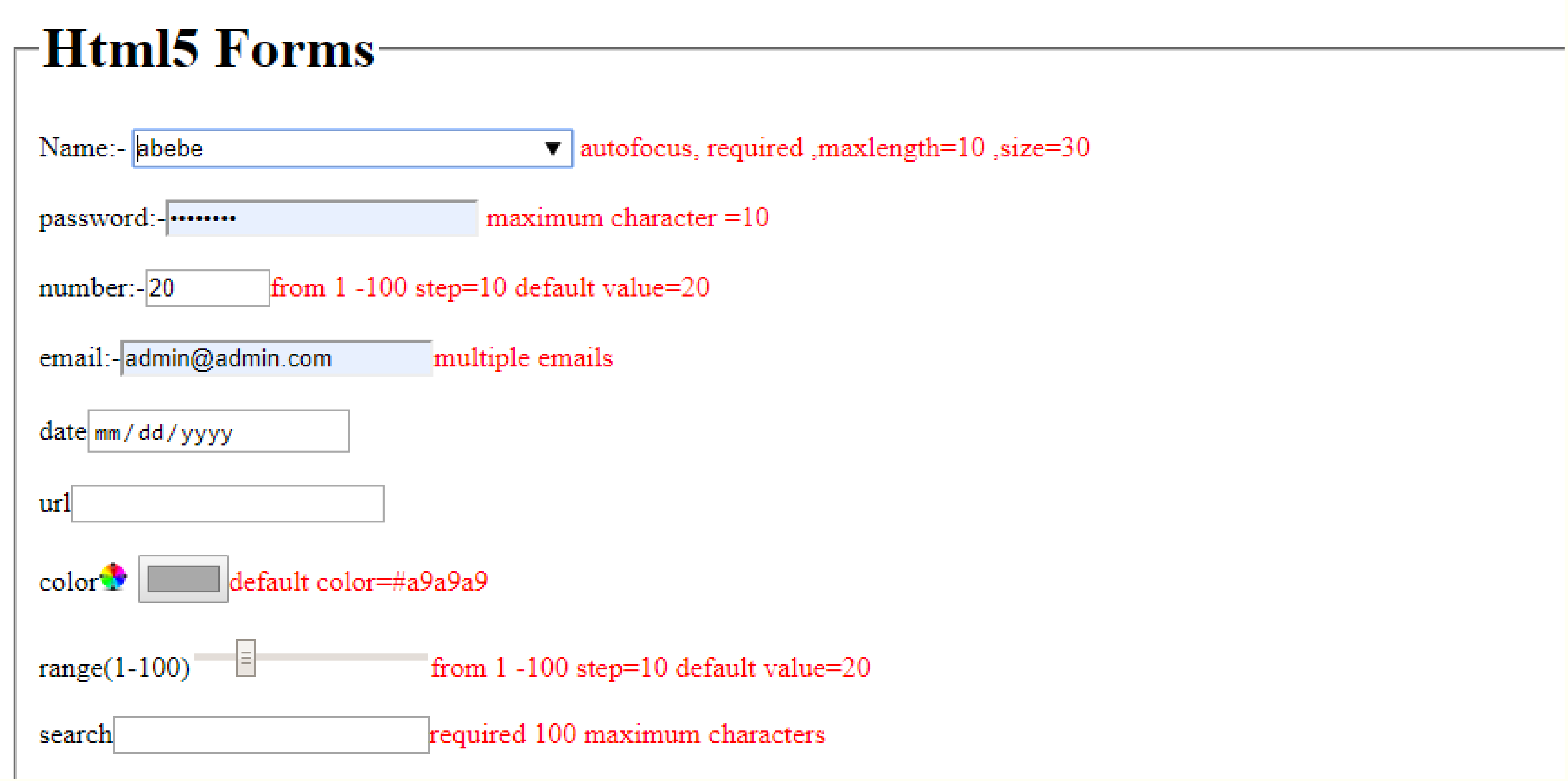
```
<p> URL: <input type="url" name="url1" /> </p>
```



HTML Input Attributes(input restrictions)

- **autofocus**: to set a controls focus on page load
- **required**: to add automatic required validation through the validation API
- **placeholder**: to add a short hint to describe the expected value of an input field
- **readonly**: to set an input field read-only
- **disabled**: to set an input as unusable and un-clickable (Won't be submitted on a form post)
- **value:** specifies the initial value for an input.
- **size**: specifies the size (in characters) for the input field
- **maxlength:** specifies the maximum allowed length

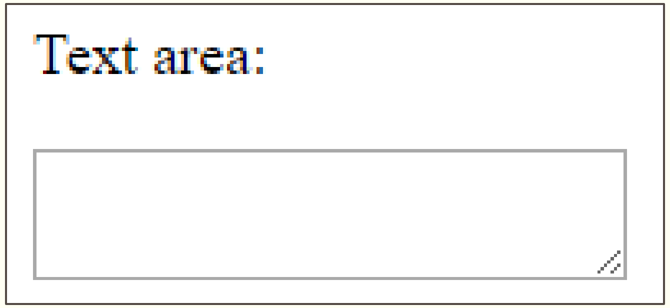
Quiz 13…input element



HTML Text Elements
> TextArea: <textarea />

- Text area fields are used when you want the user to type a multiline letters, numbers, etc. in a form.
- <textarea> defines a multi-line input field that a user can enter a multiline text into.
- Example:

```html
<p> Text area: </p>
<p> <textarea name="textarea1"></textarea> </p>
```



Quiz 14...text area element

Checkboxes and Radio Buttons

- Checkboxes and radio buttons are used to allow user to select from a set of options.
  - Checkboxes are either on or off.
  - Radio buttons are mutually exclusive controls.

Checkboxes

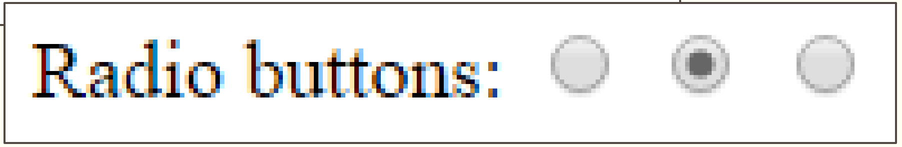
- <input type=“checkbox”> defines a checkbox. Checkboxes let a user select ZERO or MORE options of a limited number of choices.
- The **checked** attribute allows you to specify that control defaults to a checked state.
- Example:

```
<p>
    Checkboxes:
    <input type="checkbox" name="check1" checked autofocus />
    <input type="checkbox" name="check2" />
    <input type="checkbox" name="check3" checked />
</p>
```

Radio Buttons

- <input type=“radio”> defines a radio button. Radio buttons let user select ONLY ONE of a limited number of choices.
- Radio buttons are grouped together through their **name** attribute and that is how the mutual exclusive behavior is achieved.
- The **checked** attribute allows you to specify that control defaults to a checked state.

```html
<p>
    Radio buttons:
    <input type="radio" name="radio1" value="one" />
    <input type="radio" name="radio1" value="two" checked />
    <input type="radio" name="radio1" value="three" />
</p>
```

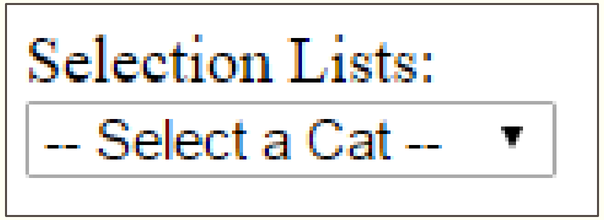


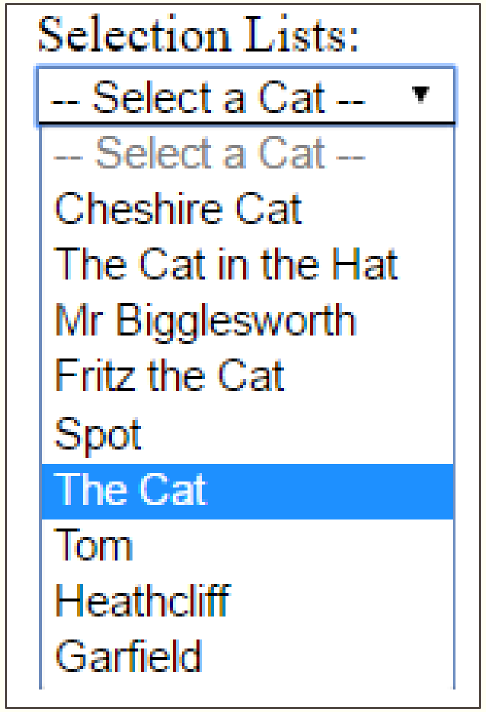
Select Lists / Dropdown Lists

- Select lists are another common feature for selecting from a list of options.
- Select lists are implemented using the <select> tag.
- The **select** tag is a container and it contains the **option** tag.
- The option tag is also a container, and it contains the text label for the dropdown list.
- The **selected** attribute on the options tag allows you in specifying the default selected option and the **disabled** attribute allows you in disabling an option from getting selected by the user.
- The **value** attribute on the option tag specifies the value for the element whose name is specified between the start and end tag of the option tag.

Select Lists / Dropdown Lists

```html
<p>
    Selection Lists: <br />
    <select name="select1">
        <option value="" disabled selected>-- Select a Cat --</option>
        <option value="cheshire">Cheshire Cat</option>
        <option value="hat">The Cat in the Hat</option>
        <option value="biggleswORTH">Mr Biggleswort</option>
        <option value="fritz">Fritz the Cat</option>
        <option value="spot">Spot</option>
        <option value="cat">The Cat</option>
        <option value="tom">Tom</option>
        <option value="heathcliff">Heathcliff</option>
        <option value="garfield">Garfield</option>
    </select>
</p>
```

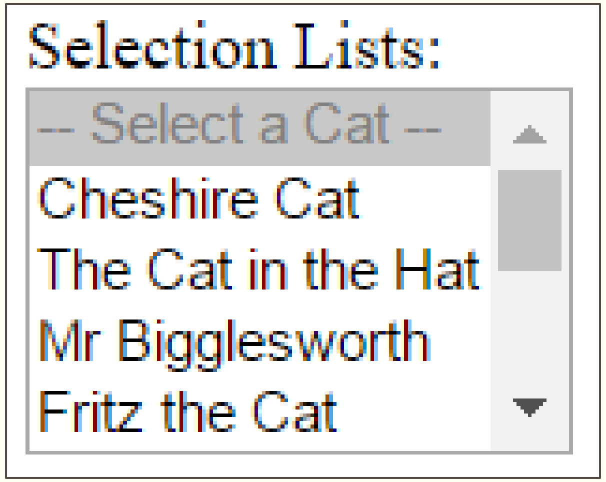




Select Lists and Dropdown Lists

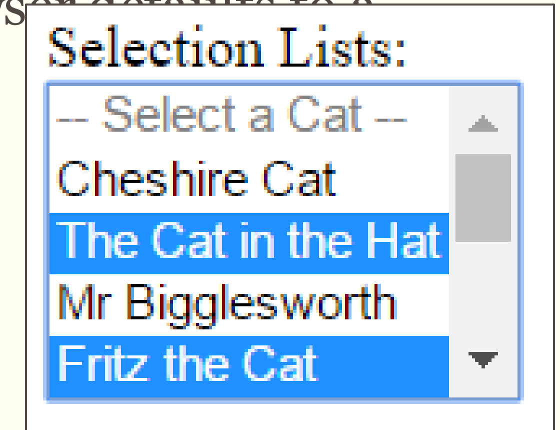
- The **size** attribute on a select list allows you to specify the number of options that are going to be displayed at the same time. The default is 1.

```
<select name="select1" size="5">
```



- The **multiple** attribute on a select list allows selecting multiple item from a list. If the multiple attribute is selected and no size attribute is specified, the browser defaults to a size of four.

```
<select name="select1" size="5" multiple>
```

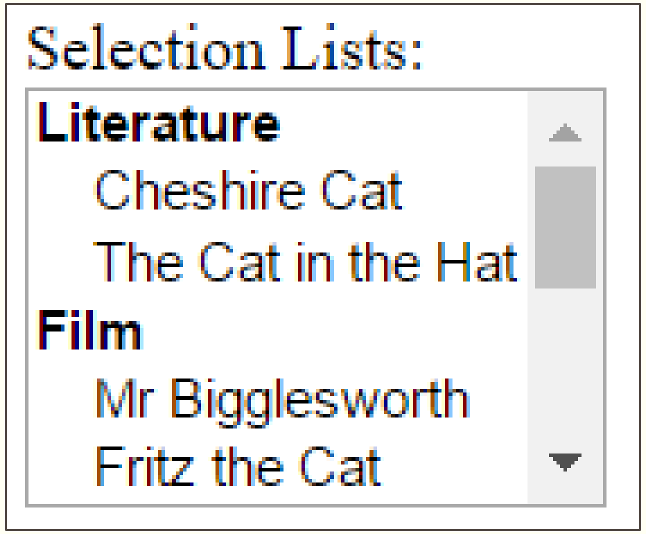


Select Lists and Dropdown Lists

- When your list options gets very long, sometimes you want to segment them. That can be achieved with the <optgroup> container tag.
- The **label** attribute on an option group allows you in specifying the name of the group.

Select Lists and Dropdown Lists

```html
<p>
    Selection Lists: <br />
    <select name="select1" size="6" multiple>
        <optgroup label="Literature">
            <option value="cheshire">Cheshire Cat</option>
            <option value="hat">The Cat in the Hat</option>
        </optgroup>
        <optgroup label="Film">
            <option value="bigglesworth">Mr Bigglesworth</option>
            <option value="fritz">Fritz the Cat</option>
        </optgroup>
```



Quiz 15…radio,checkbox and dropdown

Choose Car Types

fiat ☑
volvo ☐
vitz ☐
other ☐

Enter your Gender

◉ Male
○ Female
○ Other

Select your Education

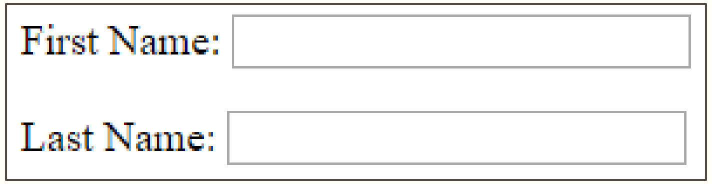
diploma ▾

diploma(disable) ,degree,masters(default),other

HTML Label

- The <label> tag defines a label for an <input> element.
- The **for** attribute of the <label> tag should be equal to the id attribute of the related element to bind them together.
  - A label can be bound to an element either by using the “for” attribute, or by placing the element inside the <label> element.

```html
<p>
    <label>
        First Name: <input type="text" name="firtName" id="firtName" />
    </label>
</p>
<p>
    <label for="lastName">Last Name: </label>
    <input type="text" name="lastName" id="lastName" />
</p>
```

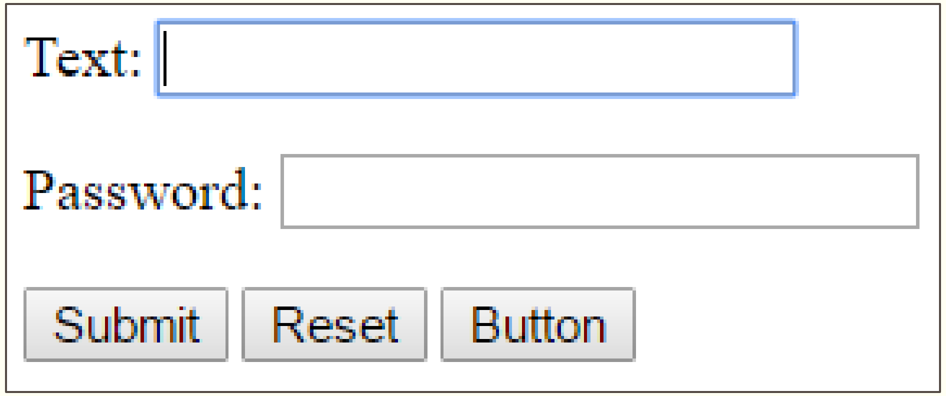


Submit and Button Elements

- Form buttons are used to initiate an action.
- The submit input button allows you to submit information on a form to a server side script.
  - <input type="submit" />
- The reset input button allows you to reset input values to their default values. (It clears the form, and resets it to the original state)
  - <input type="reset" />
- The button input button allows you to process a form submit but doesn't send a request to a server. Its only purpose is to call JavaScript.
  - Commonly used on a JavaScript only applications.
  - <input type="button />

Submit and Button Elements

```
<form id="f1" action="http://zekarias.me/test.php" onsubmit="return submitDisplay();" method="post">
    <p> Text: <input type="text" name="text1" required autofocus /> </p>
    <p> Password: <input type="password" name="password1" required /> </p>
    <p>
        <input type="submit" name="submit1" value="Submit" onclick="return submitDisplay()" />
        <input type="reset" name="reset1" value="Reset" />
        <input type="button" name="button1" value="Button" onclick="return submitDisplay()" />
    </p>
</form>
```



Quiz 16…submit,reset,image,js buttons

Form Action Attribute

- Action Attribute and the Submit Button:
  - When the user clicks on the “Submit” button, the content of the form is sent to the server.
  - The form’s action attribute defines the name of the file to send the content to.
  - It depends on PHP or ASP File or any CGI script.

```html
<form name="input" action="html_form_submit.asp" method="get">
    Username: <input type="text" name="user" />
    <input type="submit" value="Submit" />
</form>
```

Other Form Attributes

- **novalidate:** specifies that the form data should not be validated when submitted.
- **action:** specifies the URL of a file that will process the request when the form is submitted.
- **method:** defines the HTTP method for sending form-data to the action URL.

Using an Image as a submit button

- Most operating systems have good looking buttons but sometimes they don’t fit in with the design of a website.
- On such scenarios, we may use an image button that matches the design of our website.
  - **<input type=“image” src=“images/button.png” name=“submit1” value=“My Button” onclick=“return display()” />**

Hidden Element

- Problems with the web works (a problem with HTTP protocol)
  - Stateless
- HTML Forms supports an invisible element called hidden fields.
- Usage:
  - Keep track of User Id
  - Keep track of Session
- Important because they are not persistent like a session. When the page is gone, the data is gone with it.

```html
<input type="hidden" name="userid" value="12345" />
<input type="hidden" name="page" value="47" />
<input type="hidden" name="action" value="go" />
<input type="hidden" name="color" value="blue" />
```

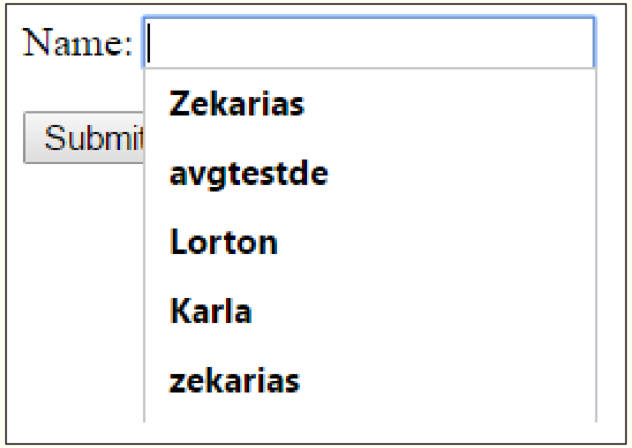
Setting Tab Order

- In HTML form, pressing the tab key on a keyboard will take you from one field to the next.
  - With shift + tab you may navigate back wards.
- By default, the tab order works on a browser on the sequence we have added the elements in the script.
- Tab order can be changed by using the **tabindex** attribute.

```
<p> Text 1: <input type="text" name="text1" autofocus tabindex="1" /></p>
<p> Text 2: <input type="text" name="text2" tabindex="2" /></p>
<p> Text 3: <input type="text" name="text3" tabindex="3" /></p>
<p> Text 4: <input type="text" name="text4" tabindex="4" /></p>
<p> <input type="submit" name="submit1" /></p>
```

Autocomplete list using the **datalist** feature

- Most browsers supports some form of auto complete for fields that you have entered on a page.
  - By default autocomplete is on.
- One may turn off auto complete by setting the **autocomplete** property of a form or an input control to **off**.
  - You may also turn **off** autocomplete at a form level and set it to **on** at an element level.



Autocomplete list using the **datalist** feature

- It is also possible to prefill your own list of autocomplete options using HTML5 new datalist feature.
- The <datalist> container element allows you to specify a list of <option> elements that can be used as a list of data for autocomplete on a browser.
- The **value** attribute on the option tag allows you to specify the value that is going to be set on the control.
  - There is also possibility of adding a text inside the option tag, but it is recommended not to use it.
- The <datalist> element must have an **id** attribute and the id should be set as the value for the **list** attribute on the input control we want to enable autocomplete on.

Autocomplete list using the **datalist** feature

```
<form method="post">

    <p> Name: <input type="text" name="FirstName" autofocus list="firstnames" /></p>
    <p> <input type="submit" name="submit1" /></p>

</form>

<datalist id="firstnames">
    <option value="Ezra"></option>
    <option value="Martha"></option>
    <option value="Tadios"></option>
    <option value="Mahelet"></option>
    <option value="Sem"></option>
</datalist>
```